MILITARY SPECIFICATION

MICROCIRCUITS, LINEAR, CMOS

NEGATIVE LOGIC, ANALOG SWITCH,

MONOLITHIC SILICON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, CMOS negative logic analog switches. Three product assurance classes and a choice of case outline/lead finish are provided for each type and are reflected in the complete part number.

1.2 Part number. The part number shall be in accordance with MIL-M-38510.

1.2.1 Device type. The device type shall be as follows:

Device type	Circuit
01	Dual SPST switch
02	Quad SPST switch

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outline. The case outline shall be designated as follows:

Letter	Case outline (MIL-M-38510, appendix C)
C	D-1 (14-lead, 1/4" x 3/4"), dual-in-line package
E	D-2 (16-lead, 1/4" x 7/8"), dual-in-line package
I	A-2 (10-lead can)

1.3 Absolute maximum ratings. 1/

Supply voltage between V+ and V-	
Device types 01, 02	+40 V
Digital input voltage	$(+V_{supply}) \geq V_{IN} \geq (-0.3\ V)$
Analog input voltage	$(+V_{supply} + 2\ V) \geq V_A \geq (-V_{supply} - 2V)$
Storage temperature range	-65°C to +150°C
Lead temperature (soldering, 10 seconds)	300°C
Junction temperature (T_J)	175°C
Maximum current:	
Any terminal (except S or D)	30 mA
S or D terminals	20 mA

1/ As with all semiconductors, stresses listed under "Absolute maximum ratings" may be applied to devices (one at a time) without resulting in permanent damage. This is a stress rating only. Exposure to absolute maximum rating conditions for extended periods may affect device reliability. The conditions listed under "Electrical performance characteristics" (table I) are the only conditions recommended for satisfactory operation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: George C. Marshall Space Flight Center, National Aeronautics and Space Administration, ATTN: EG02, Marshall Space Flight Center, AL 35812, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.4 Recommended operating conditions.

Positive supply voltage	+15 V
Negative supply voltage	-15 V
Reference voltage	Open
Input (address) high voltage	2.4 V
V_{IL}	0.8 V
Ambient operating temperature range	-55°C to +125°C

1.5 Power and thermal characteristics.

Package	Case outline	Maximum allowable 2/ power dissipation	Maximum θ_{JC}	Maximum θ_{JA}
Dual-in-line	C, E	400 mW @ T_A = 125°C	35°C/W	120°C/W
10-lead can	I	350 mW @ T_A = 125°C	40°C/W	140°C/W

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARDS

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Detail specifications. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510, and herein.

3.2.1 Terminal connections. The terminal connections shall be as specified on figure 1.

3.2.2 Truth table. The truth table shall be as specified on figure 2.

3.2.3 Schematic circuits. The schematic circuits shall be as specified on figure 3.

3.2.4 Case outlines. The case outlines shall be as specified in 1.2.3.

3.2.5 Package and sealing material. Package and sealing material shall be in accordance with MIL-M-38510.

2/ All leads welded or soldered to PC board.

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TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions V^+ = 15 V, GND = 0 V, V^- = -15 V unless otherwise specified		Device type	Limits Min	Limits Max	Unit
Switch "ON" resistance (figure 7)	R_{DS}	V_{IN} = 0.8 V, V_S = 10 V, I_D = -1 mA	T_A = -55°C, +25°C	01		70	Ω
			T_A = 125°C			100	Ω
			T_A = -55°C, +25°C	02		175	Ω
			T_A = 125°C			250	Ω
		V^+ = 10 V, V^- = -10 V, V_{IN} = 0.8 V, V_S = 7.5 V, I_D = -1 mA	T_A = -55°C, +25°C	01		100	Ω
			T_A = 125°C			150	Ω
			T_A = -55°C, +25°C	02		200	Ω
			T_A = 125°C			250	Ω
		V_{IN} = 0.8 V, V_S = -10 V, I_D = 1 mA	T_A = -55°C, +25°C	01		70	Ω
			T_A = 125°C			100	Ω
			T_A = -55°C, +25°C	02		175	Ω
			T_A = 125°C			250	Ω
		V^+ = 10 V, V^- = -10 V, V_{IN} = 0.8 V, V_S = -7.5 V, I_D = 1 mA	T_A = -55°C, +25°C	01		100	Ω
			T_A = 125°C			150	Ω
			T_A = -55°C, +25°C	02		200	Ω
			T_A = 125°C			250	Ω
Source "OFF" leakage current (figure 8)	$I_{S(OFF)}$	V_S = 14 V, V_D = -14 V, V_{IN} = 2.4 V	T_A = 25°C	01,02	-2	2	nA
			T_A = 125°C		-100	100	nA
			T_A = -55°C		-100	100	nA
		V_S = -14 V, V_D = 14 V, V_{IH} = 2.4 V	T_A = 25°C	01,02	-2	2	nA
			T_A = 125°C		-100	100	nA
			T_A = -55°C		-100	100	nA
Drain "OFF" leakage current (figure 9)	$I_{D(OFF)}$	V_D = -14 V, V_S = 14 V, V_{IN} = 2.4 V	T_A = 25°C	01,02	-2	2	nA
			T_A = 125°C		-100	100	nA
			T_A = -55°C		-100	100	nA
		V_D = 14 V, V_S = -14 V, V_{IN} = 2.4 V	T_A = 25°C	01,02	-2	2	nA
			T_A = 125°C		-100	100	nA
			T_A = -55°C		-100	100	mA

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions V^+ = 15 V, GND = 0 V, V^- = -15 V unless otherwise specified		Device type	Limits		Unit
					Min	Max	
Channel "ON" leakage current (figure 10)	$I_{D(ON)}$	$V_D = V_S$ = 14 V, V_{IN} = 0.8 V	T_A = 25°C	01	-2	2	nA
				02	-2	2	nA
			T_A = 125°C	01,02	-200	200	nA
			T_A = -55°C		-200	200	nA
		$V_D = V_S$ = -14 V, V_{IN} = 0.8 V	T_A = 25°C	01	-2	2	nA
				02	-2	2	nA
			T_A = 125°C	01,02	-200	200	nA
			T_A = -55°C		-200	200	nA
Low level input current (figure 11)	I_{IL} 1/	V_{IL} = 0.8 V, V_{IH} = 2.4 V	T_A = 25°C	01,02	-0.5	0.5	µA
			T_A = 125°C		-1.0	1.0	µA
			T_A = -55°C		-1.0	1.0	µA
High level input current (figure 11)	I_{IH}	V_{IL} = 0.8 V, V_{IN} = 15 V	T_A = 25°C	01,02	-0.5	0.5	µA
			T_A = 125°C		-1.0	1.0	µA
			T_A = -55°C		-1.0	1.0	µA
Supply current (figure 12)	$+I_{CC}$	V_{IL} = 0 V (all inputs)	T_A = 25°C, 125°C	01,02		1.5	mA
			T_A = -55°C			2.0	mA
		V_{IH} = 5 V (all inputs)	T_A = 25°C, 125°C	01,02		1.5	mA
			T_A = -55°C			2.0	mA
Supply current (figure 12)	$-I_{CC}$	V_{IL} = 0 V (all inputs)	T_A = 25°C, 125°C	01,02	-1.5		mA
			T_A = -55°C		-2.0		mA
		V_{IH} = 5 V (all inputs)	T_A = 25°C, 125°C	01,02	-1.5		mA
			T_A = -55°C		-2.0		mA
Capacitance: address	C_A	T_A = 25°C, GND = 0 V, f = 1 MHz (see 4.4.1c)	V_{IL} = 0 V	01,02		15	pF
Capacitance: input switch	C_{IS}	T_A = 25°C, GND = 0 V, f = 1 MHz (see 4.4.1c)	V_{IH} = 5 V	01,02		15	pF

1/ Input current at one input node.

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions V^+ = 15 V, GND = 0 V, V^- = -15 V unless otherwise specified		Device type	Limits		Unit
					Min	Max	
Capacitance: output switch	C_{OS}	T_A = 25°C GND = 0 V f = 1 MHz (see 4.4.1c)	V_{IH} = 5 V	01,02		20	pF
Off isolation	V_{ISO}	T_A = 25°C, f = 200 kHz, V_{gen} = 1 vp-p (see 4.4.4b)		01,02	60		dB
Crosstalk between channels	V_{CT}	T_A = 25°C, f = 200 kHz, V_{gen} = 1 vp-p (see 4.4.4b)		01,02	60		dB
Charge transfer error	V_{CTE}	T_A = 25°C (see 4.4.4b)		01,02	-10	10	mV
Input test voltage	V_{ZAP}	C_1 = 100 pF, R_2 = 1.5 kΩ (see 4.5.3)		01,02	400		V
Turn "ON" time	$t_{(ON)}$	C_L = 100 pF R_L = 1 kΩ (Figure 13)	T_A = 125°C	01,02		800	ns
			T_A = -55°C, 25°C			600	ns
Turn "OFF" time	$t_{(OFF)}$	C_L = 100 pF R_L = 1 kΩ (Figure 13)	T_A = 125°C	01,02		650	ns
			T_A = -55°C, 25°C			500	ns

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3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510 (see 6.5).

3.4 Electrical performance characteristics. The electrical performance characteristics are as specified in table I, and apply over the full recommended ambient operating temperature range, unless otherwise specified.

3.5 Electrical test requirements. The electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III.

3.6 Marking. Marking shall be in accordance with MIL-M-38510. At the option of the manufacturer, marking of the country of origin may be omitted from the body of the microcircuit, but shall be retained on the initial container.

3.6.1 Serialization. All class S devices shall be serialized in accordance with MIL-M-38510.

3.6.2 Correctness of indexing and markings. All devices shall be subjected to the final electrical tests specified in table II after part marking to verify that they are correctly indexed and identified by part number. Optionally, an approved electrical test may be devised especially for this requirement.

3.7 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 57 (see MIL-M-38510, appendix E).

TABLE II. Electrical test requirements.

MIL-STD-883 test requirements	Subgroups (see table III) 3/		
	Class S 1/ 2/	Class B 1/ 2/	Class C 2/
Interim electrical parameters (pre-burn-in) method 5004	1	1	None
Final electrical parameters method 5004	1*,2,3,9	1*,2,3,9	1
Group A test requirements method 5005 3/	1,2,3,4,9,10,11	1,2,3,4,9,10,11	1,2,3,4,9
Group B V_{ZAP} requirements	see 4.4.2 and 4.5.3	see 4.4.2 and 4.5.3	N/A
Group C end-point and group B class S, electrical parameters method 5005	1,2,3, and table IV delta limits	1 and table IV delta limits	1 and table IV delta limits
Additional electrical subgroups for group C periodic inspection	N/A	10, 11	10, 11
Group D end-point electrical parameters method 5005	1, 2, 3, **(12,13,14)	1 **(12,13,14)	1 **(12,13,14)

1/ (*) indicates PDA applies to subgroup 1 (see 4.2d).

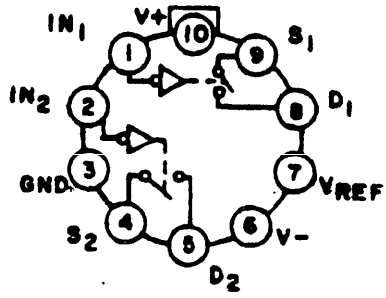
2/ (**) See paragraph 4.4.4b.

3/ Subgroup 4, see 4.4.1c.

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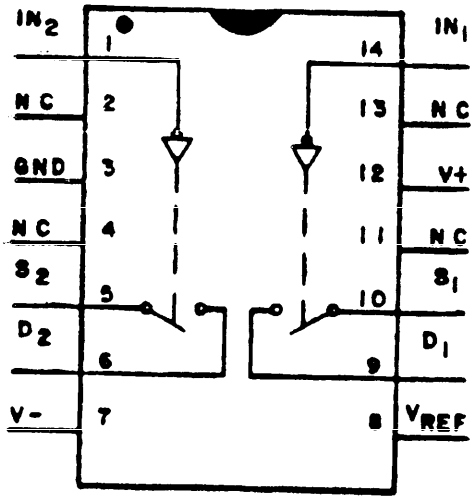
Device type 01

Case 1



Device type 01

Case C



Device type 02

Case E

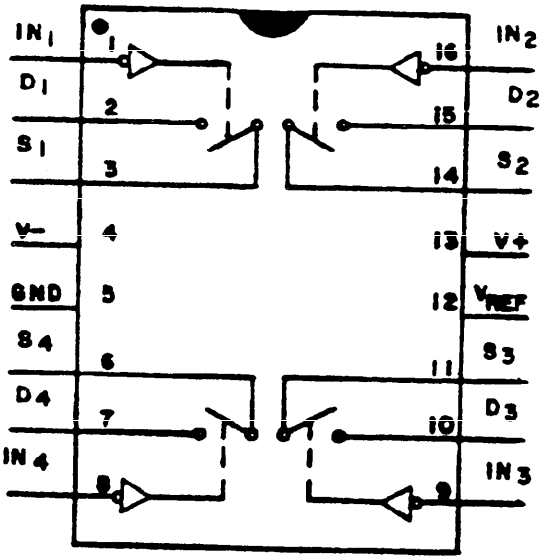


FIGURE 1. Terminal connections.

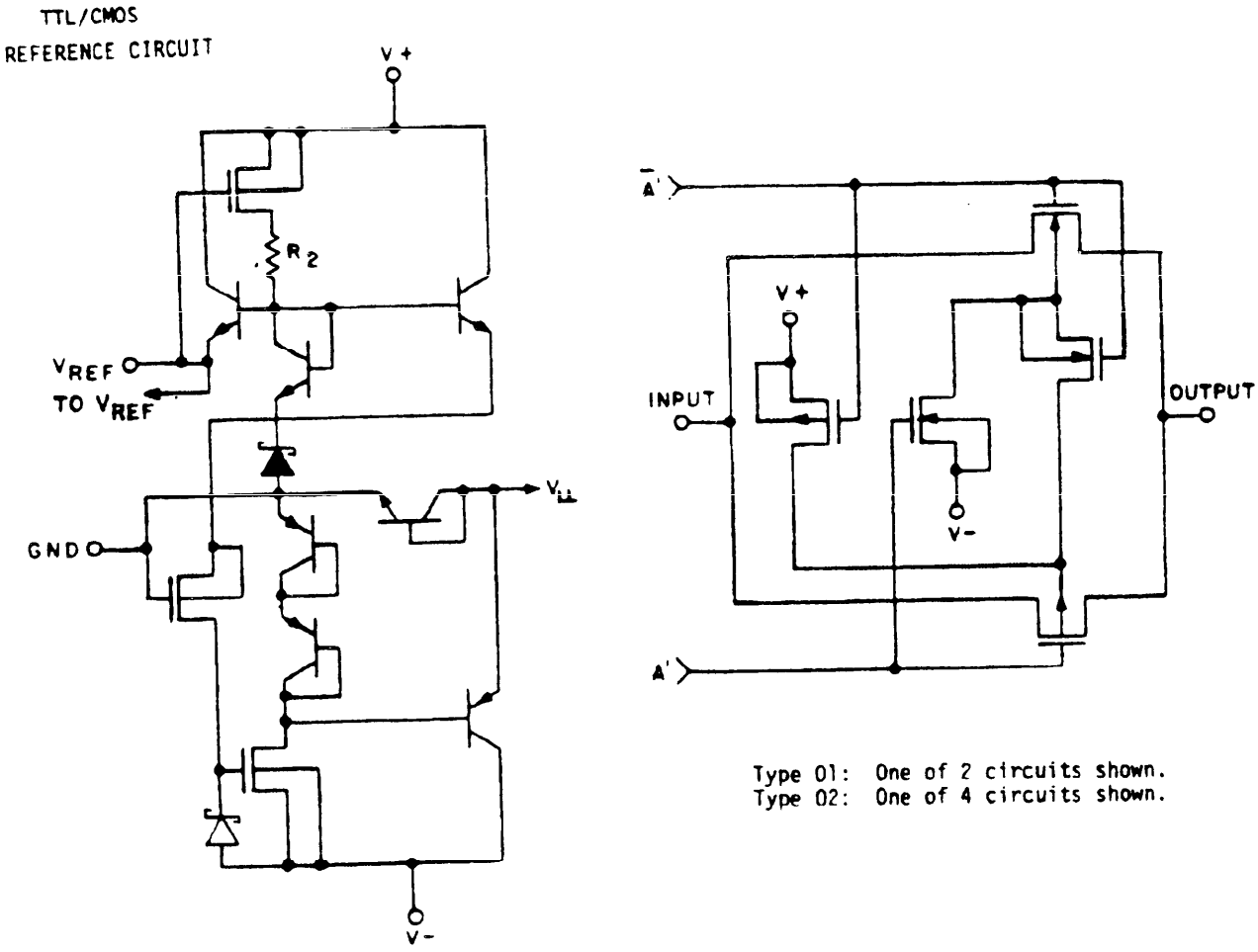
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3.2.1 Terminal connections. The terminal connections shall be as specified on

Truth table for each switch	
Input	Switch condition
$2.4V \leq V_{IN} \leq 15.0V$	OPEN
$0V \leq V_{IN} \leq 0.8$ V	Closed

FIGURE 2. Truth table (all device types).

Device types 01 and 02



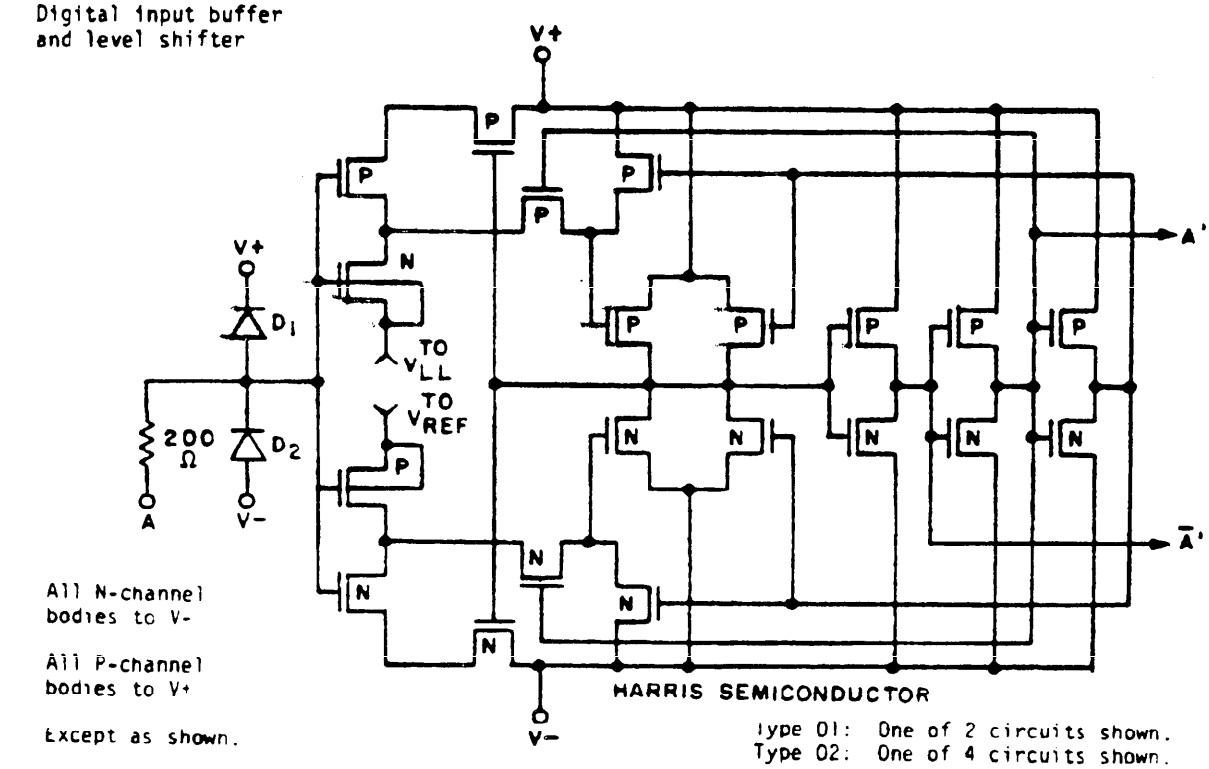
Type 01: One of 2 circuits shown.
Type 02: One of 4 circuits shown.

FIGURE 3. Schematic circuits.

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Device types 01 and 02

Circuit A



Device types 01 and 02

Circuit B

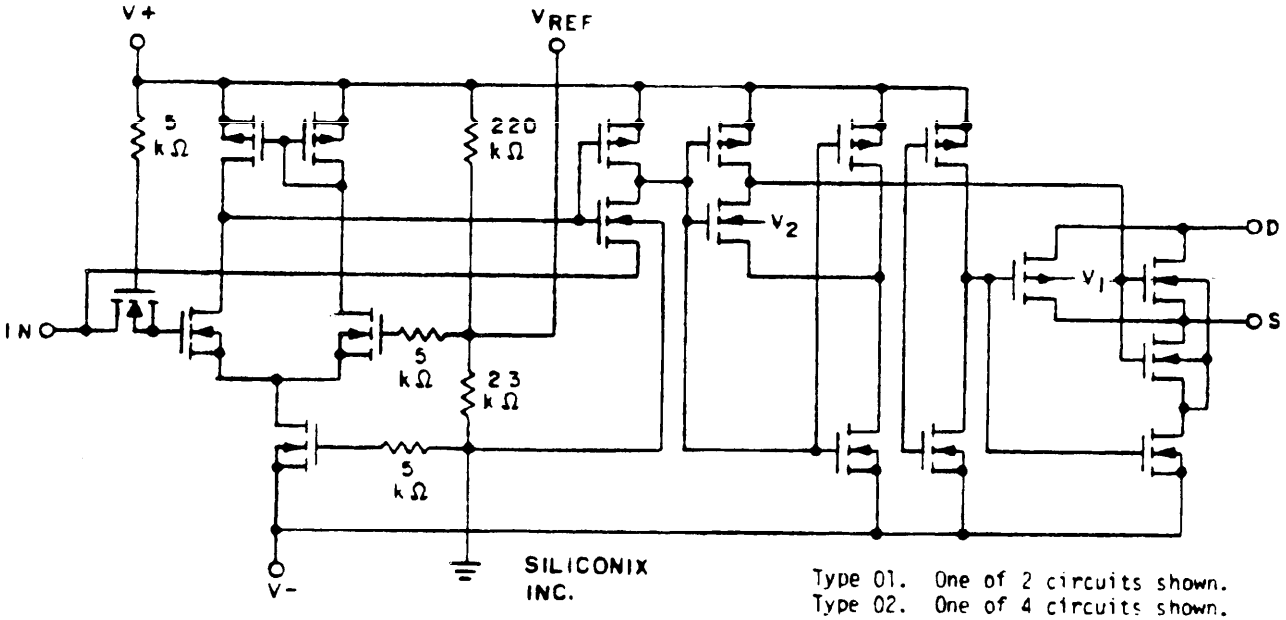
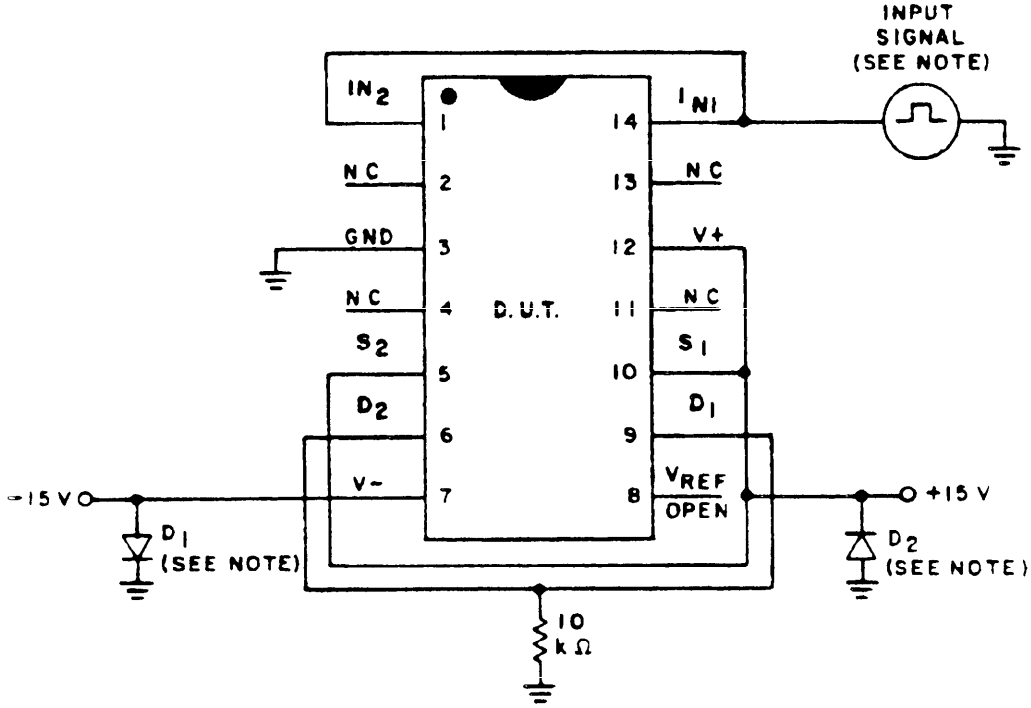


FIGURE 3. Schematic circuits - Continued.

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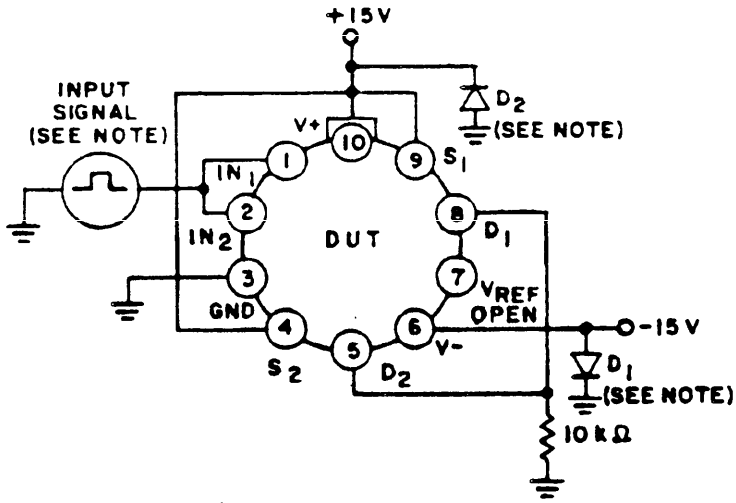
Device type 01

Case C



Device type 01

Case I



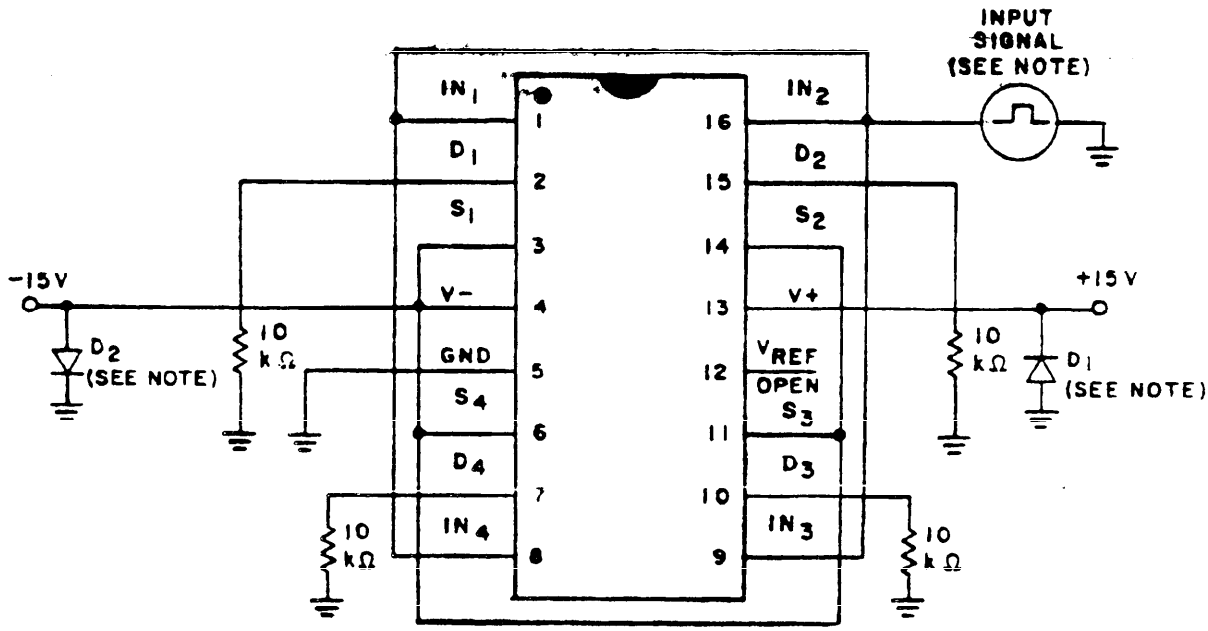
NOTE: Input signal requirements:
a. Square wave, 50% duty cycle.
b. 25 kHz ≤ PRR ≤ 1 MHz.
c. t_{TLH} and t_{THL} ≤ 1 µs.
d. Voltage = ±15 V.
e. Diodes (D1 and D2) = 1N4002 or equivalent.

FIGURE 4. Dynamic burn-in and steady state life test circuits.

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Device type 02

Case E

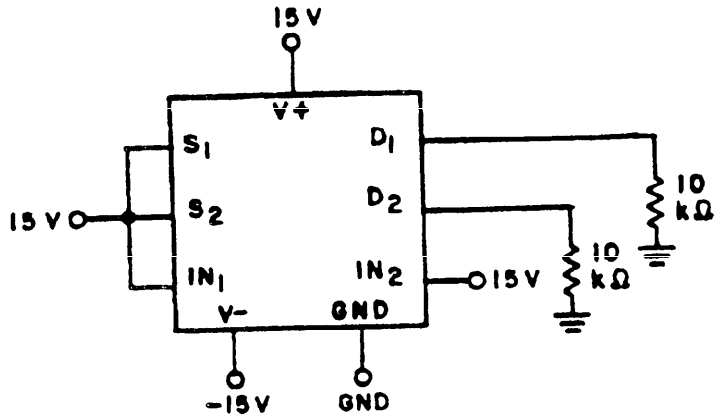


NOTE: Input signal requirements:
a. Square wave, 50% duty cycle.
b. 25 kHz ≤ PRR ≤ 1 MHz.
c. t_{TLH} and t_{THL} ≤ 1 µs.
d. Voltage = ±15 V.
e. Diodes (D1 and D2) = 1N4002 or equivalent.

FIGURE 4. Dynamic burn-in and steady state life test circuits - Continued.

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Device type 01



Device type 02

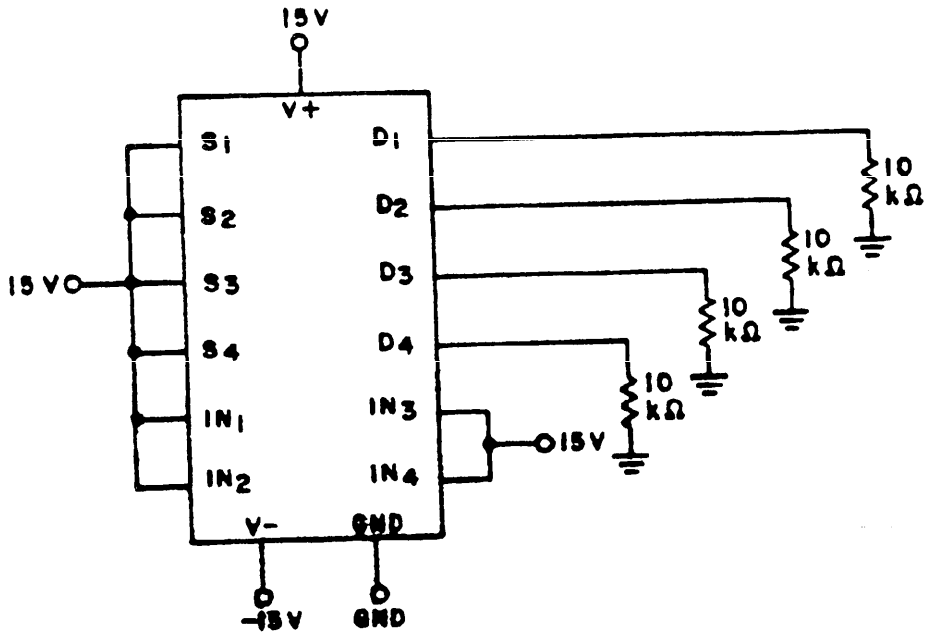
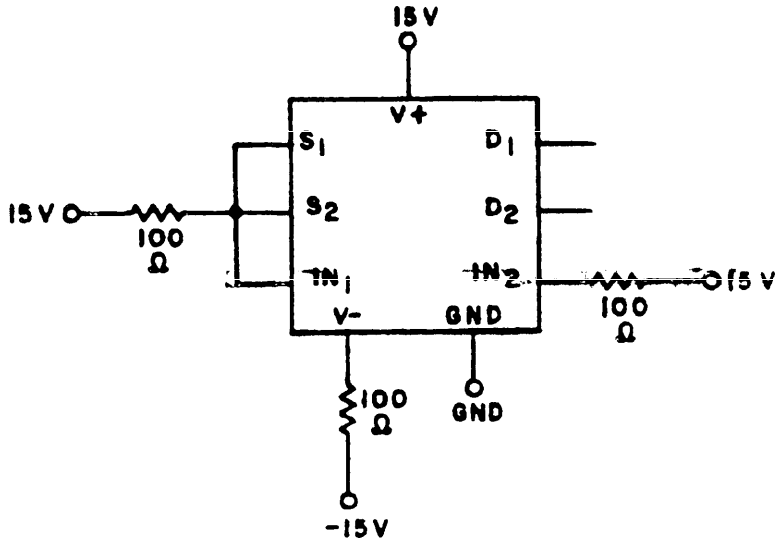


FIGURE 5. Test circuits, burn-in (steady state power and reverse bias) and life test.

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Device type 01



Device type 02

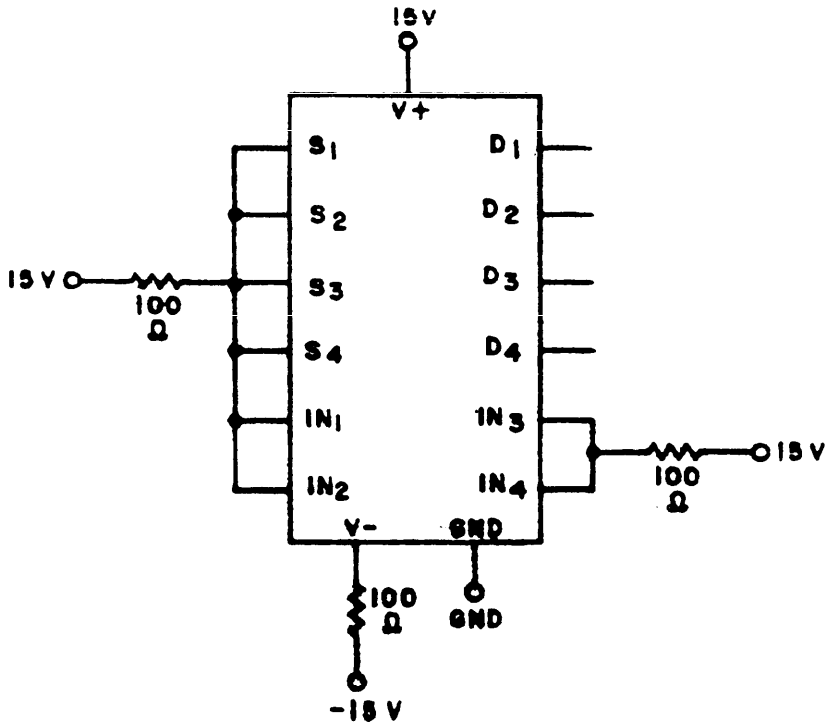
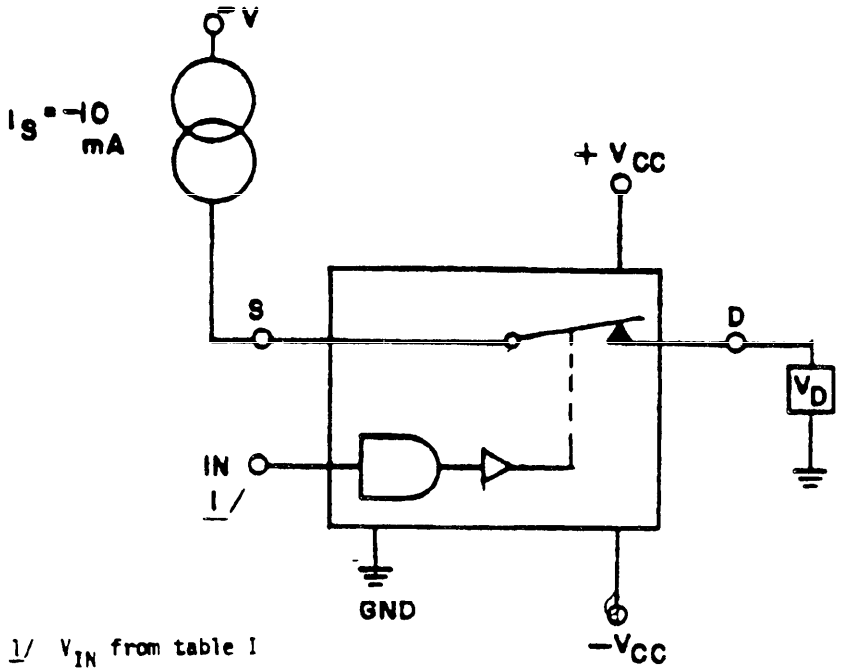


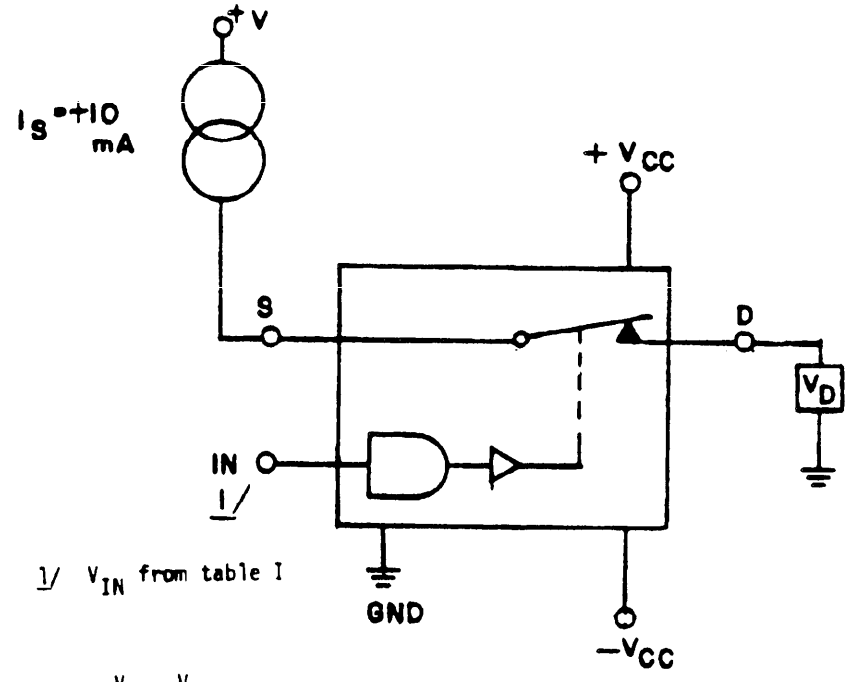
FIGURE 6. Test circuits, accelerated burn-in and life test.

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1/ V_{IN} from table I

$$R_{DS} = \frac{V_S - V_D}{-10\ mA}$$

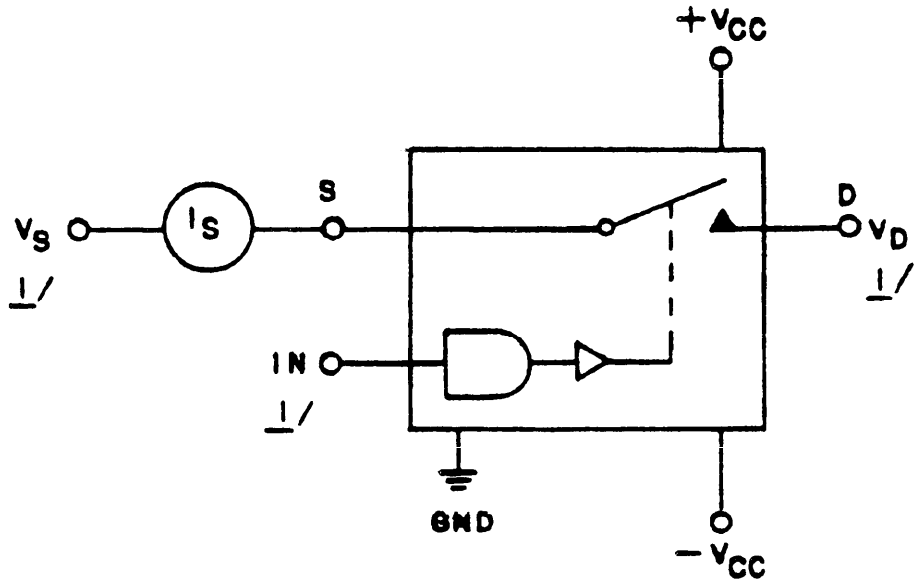


1/ V_{IN} from table I

$$R_{DS} = \frac{V_S - V_D}{10\ mA}$$

FIGURE 7. R_{DS} test circuits.

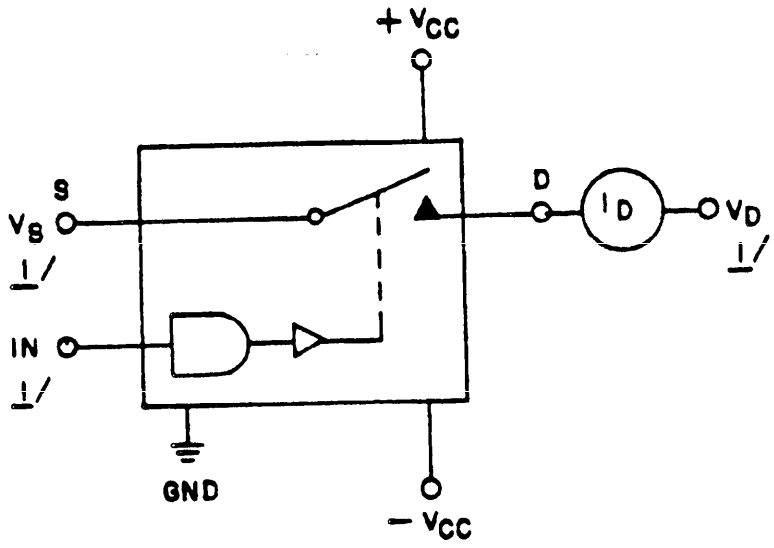
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1/ Test conditions are from table I.

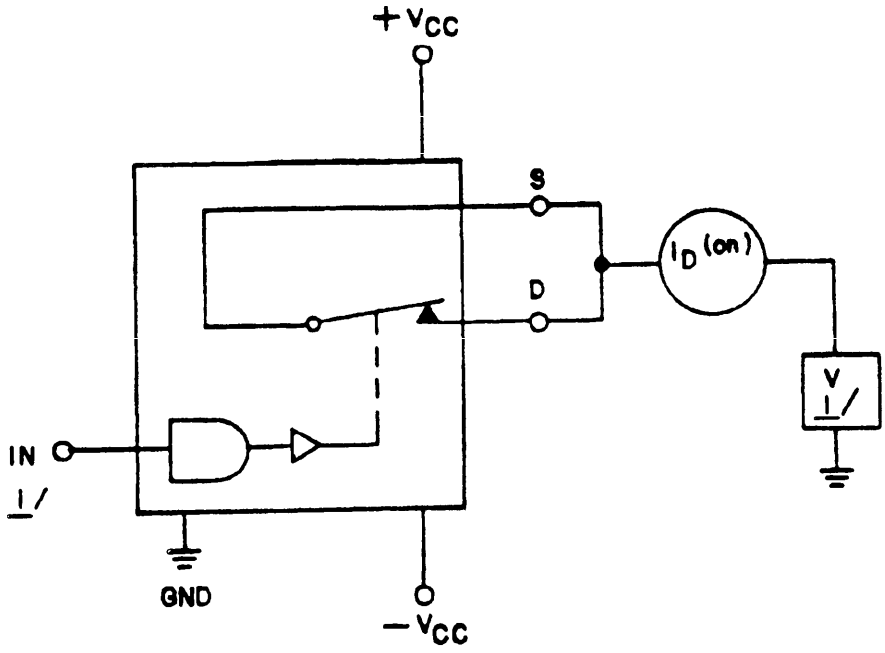
FIGURE 8. $I_{S(OFF}$ test circuit.

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1/ Test conditions are from table I.

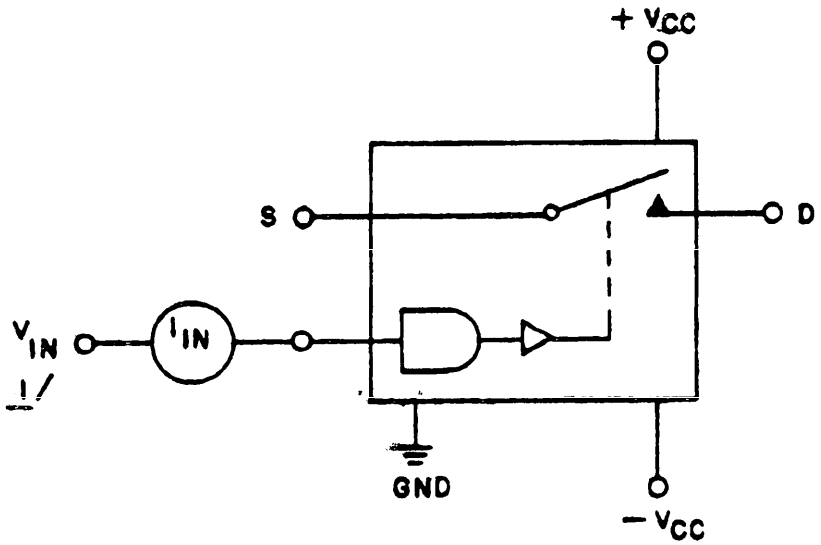
FIGURE 9. $I_{D(OFF)}$ test circuit.



1/ Conditions are from table I.

FIGURE 10. $I_{D(ON)}$ test circuit.

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1/ Test conditions are from table I.

FIGURE 11. I_{IL}, I_{IH} test circuit.

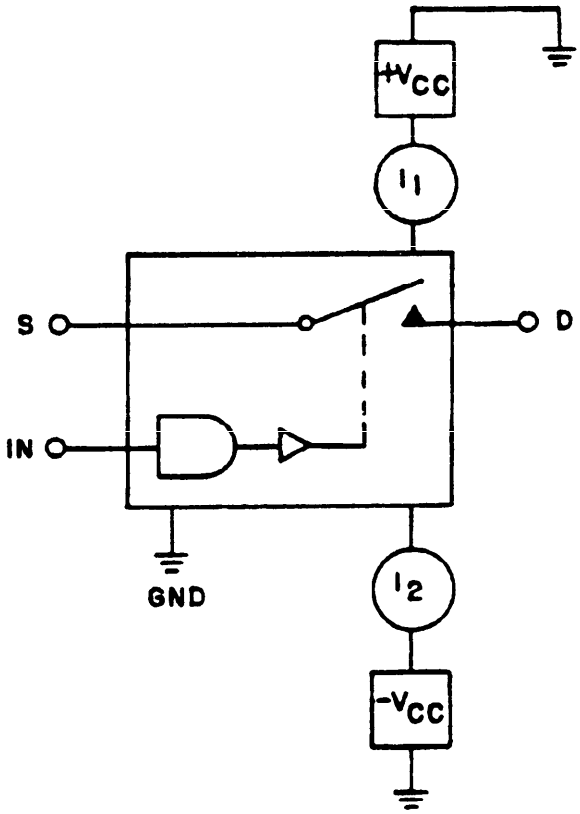
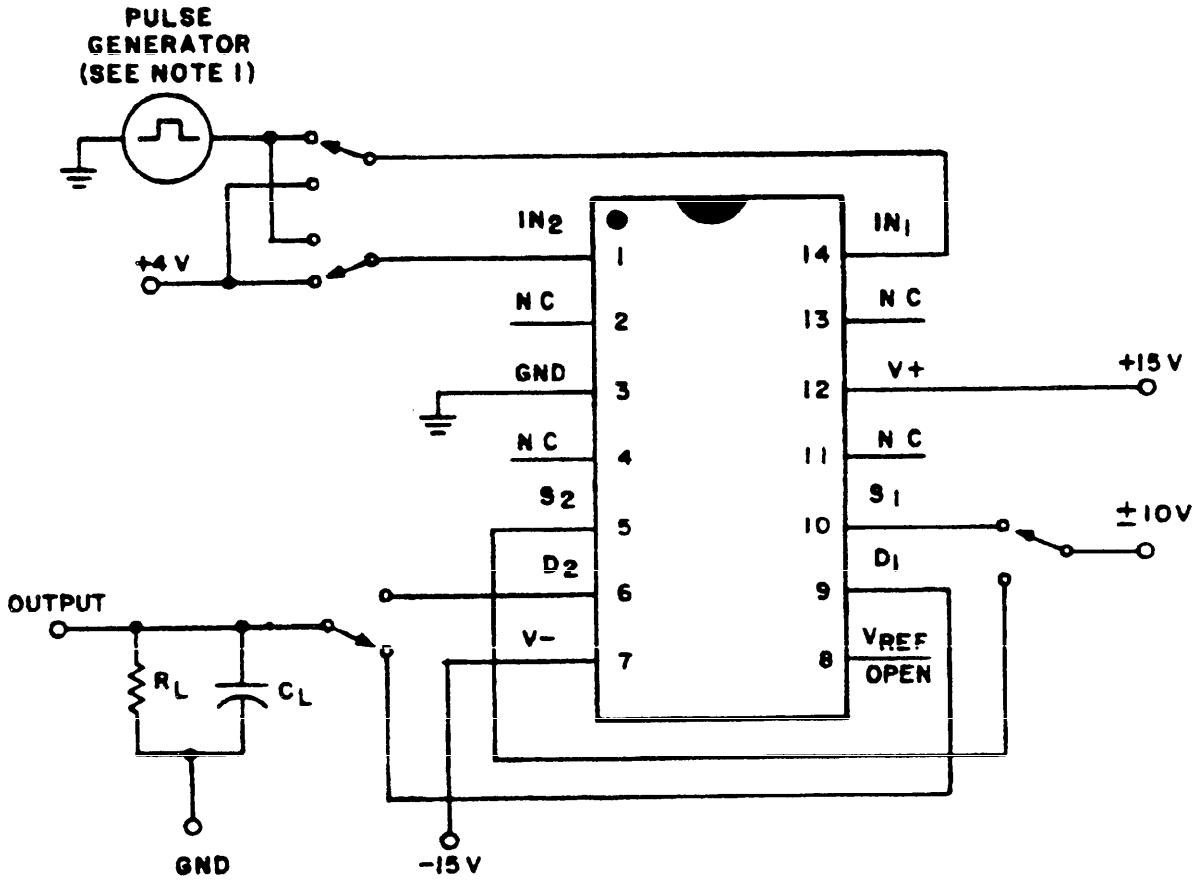


FIGURE 12. $\pm I_{CC}$ test circuit.

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Device type 01

Case C



TEST CIRCUIT

R_L = 1 kΩ ±5%.
C_L = 100 pF ±5%.
(Includes wiring and probe capacitance.)

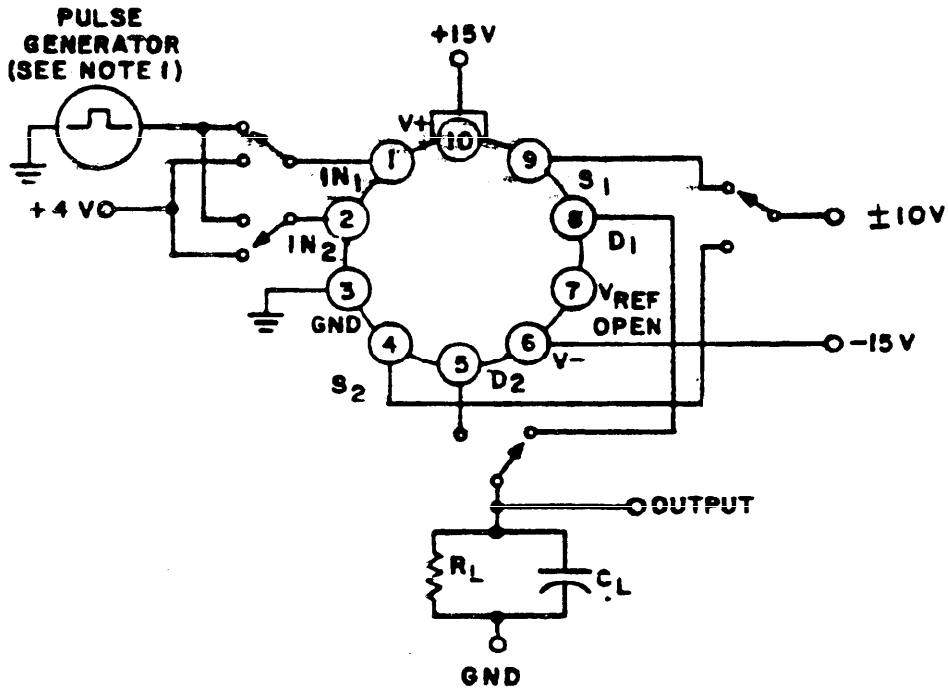
NOTES:
1. The pulse generator has the following characteristics: V_{GEN} = 4 V, t_{THL} and $t_{TLH} \leq$ 20 ns.
2. See table III for complete terminal conditions.

FIGURE 13. Switching time test circuits and waveforms.

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Device type 01

Case I



TEST CIRCUIT

R_L = 1 kΩ ±5%.
C_L = 100 pF ±5%
(Includes wiring and probe capacitance.)

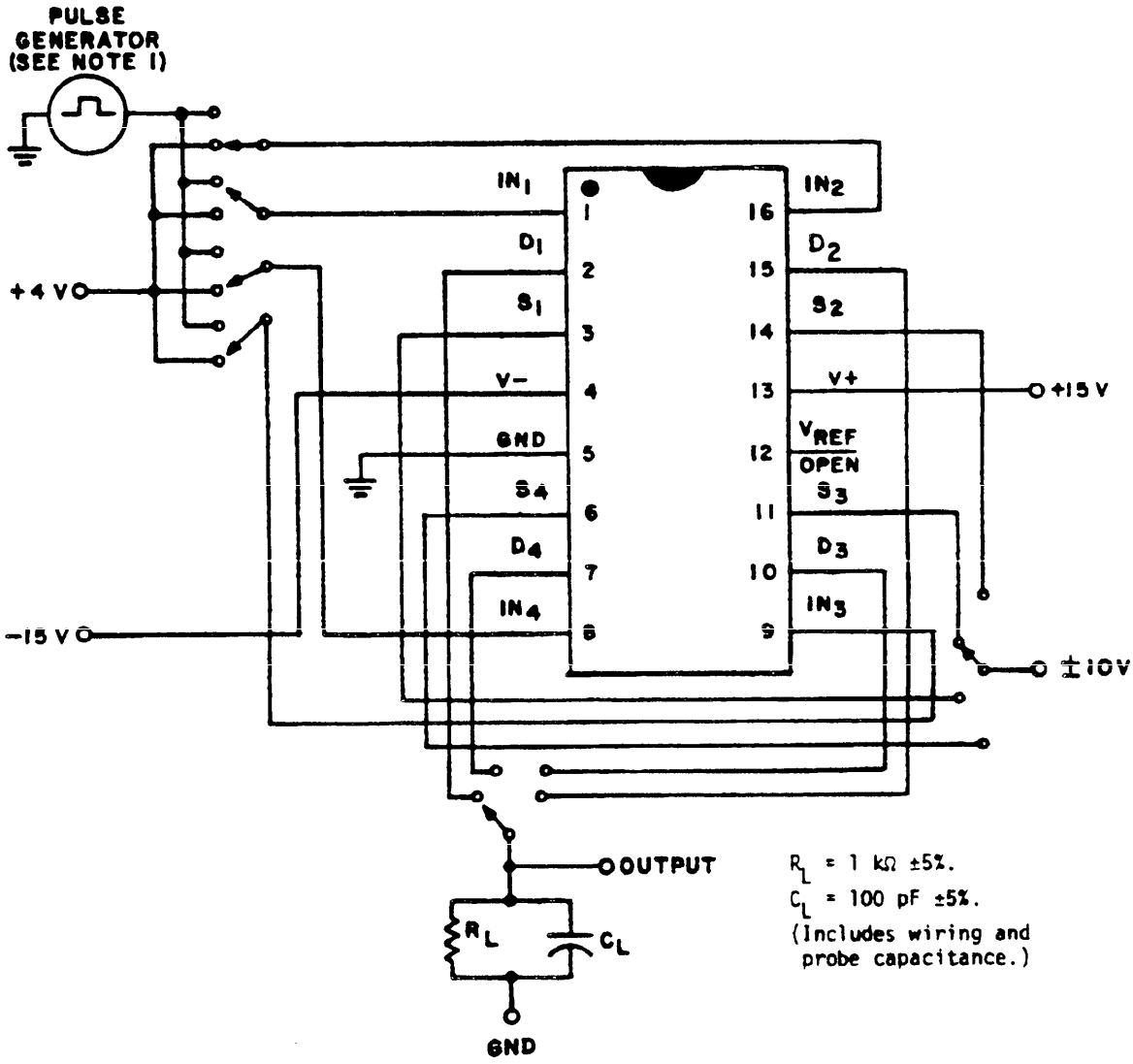
NOTES:
1. The pulse generator has the following characteristics: V_{GEN} = 4 V, t_{THL} and $t_{TLH} \leq$ 20 ns.
2. See table III for complete terminal conditions.

FIGURE 13. Switching time test circuits and waveforms - Continued.

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Device type 02

Case E



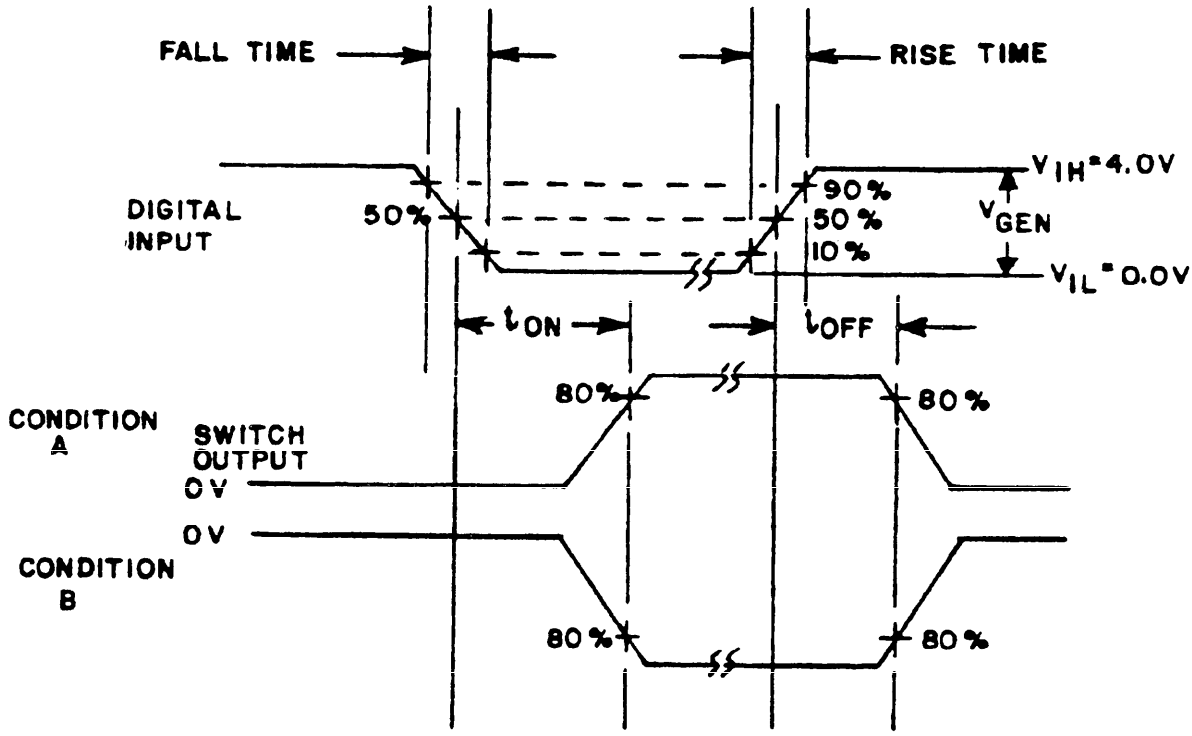
R_L = 1 kΩ ±5%.
C_L = 100 pF ±5%.
(Includes wiring and probe capacitance.)

NOTES:
1. The pulse generator has the following characteristics: V_{GEN} = 4 V, $t_{THL} \leq$ 20 ns.
2. See table III for complete terminal conditions.

FIGURE 13. Switching time test circuits and waveforms - Continued.

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Device types 01 and 02

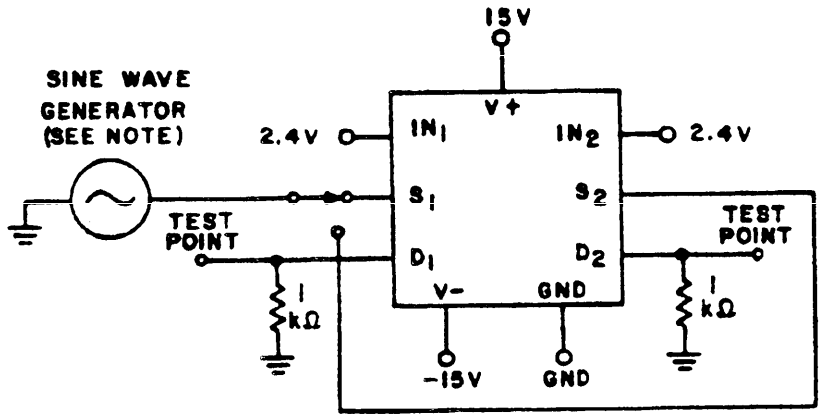


NOTE: Rise time and fall time ≤20 ns.

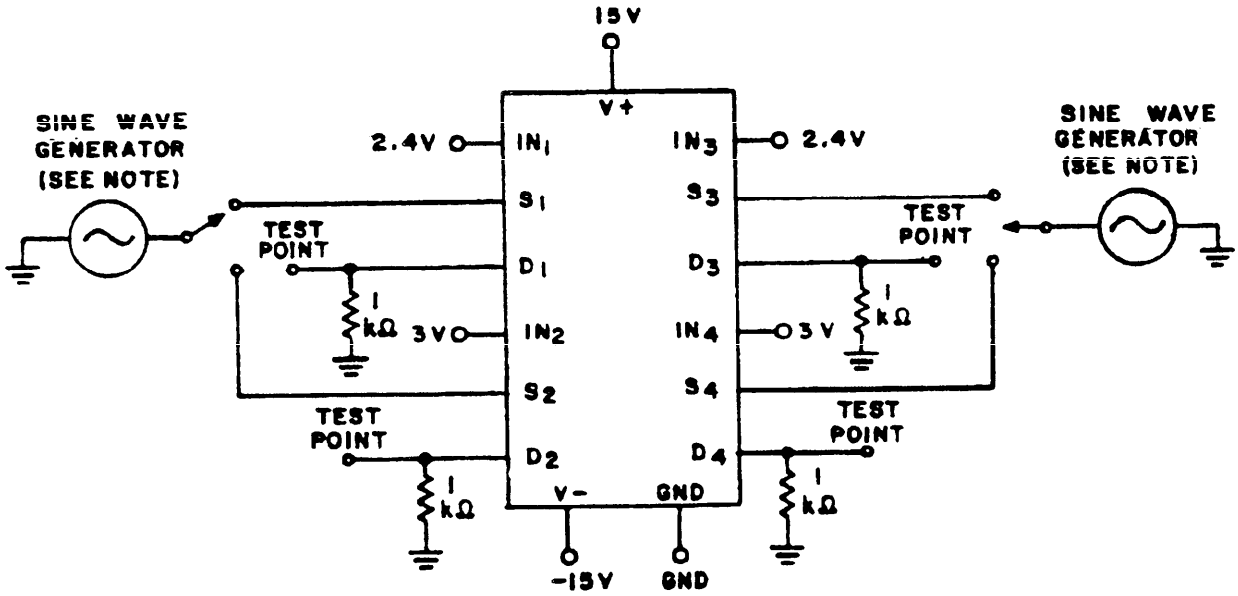
FIGURE 13. Switching time test circuits and waveforms - Continued.

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Device type 01



Device type 02

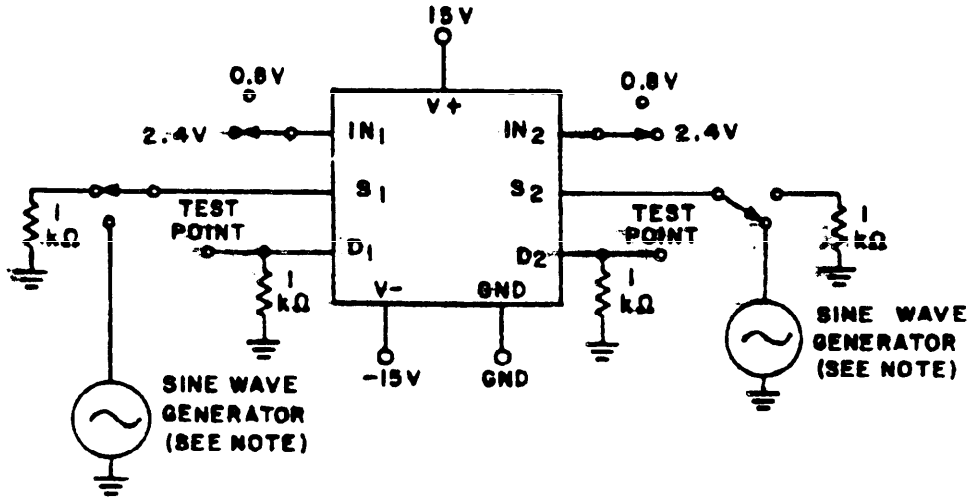


NOTE: The sine wave generator has the following characteristics:
V_{GEN} = 1 V p-p, frequency = 200 kHz.

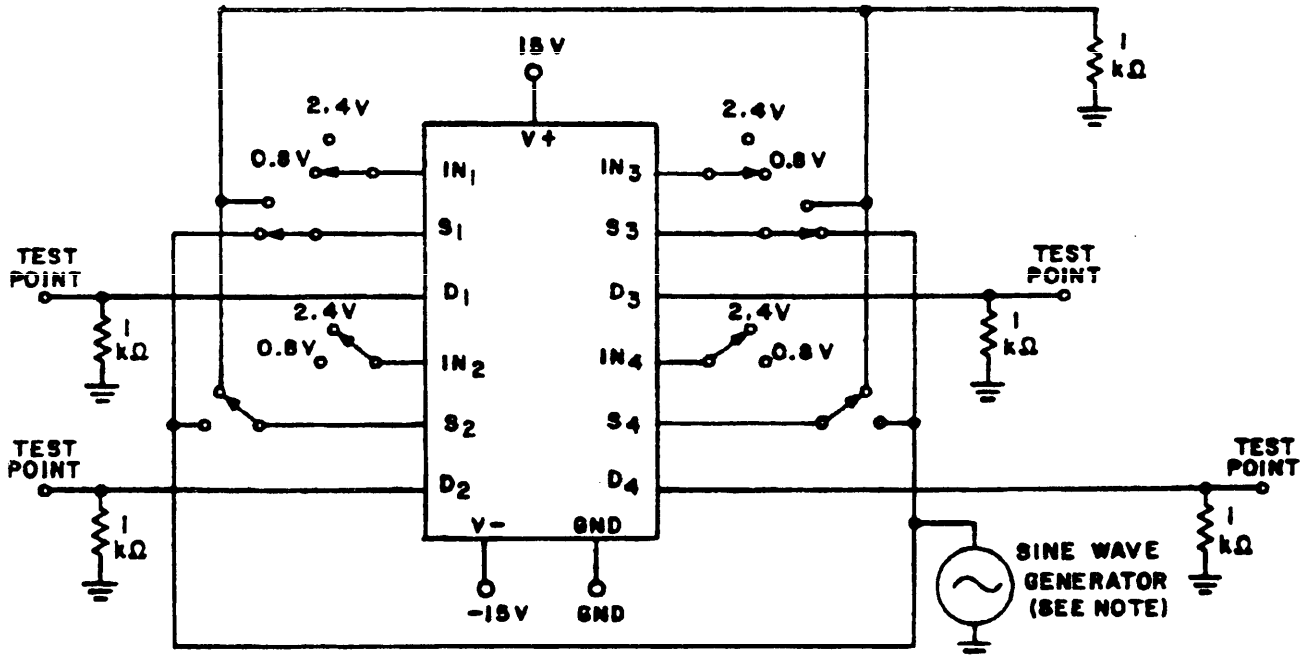
FIGURE 14. Off isolation test circuits.

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Device type 01



Device type 02

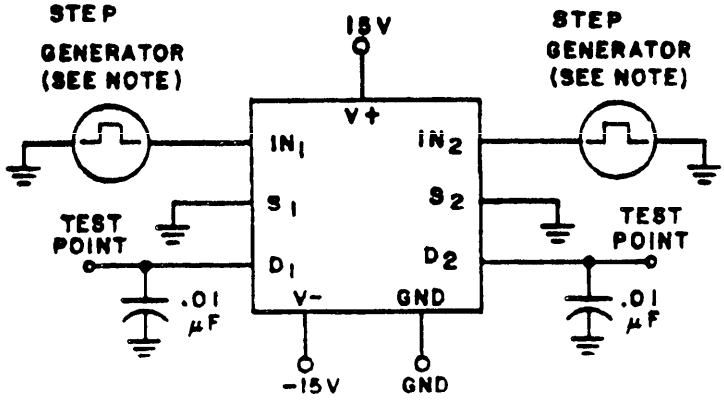


NOTE: The pulse generator has the following characteristics: V_{GEN} = 1 V p-p, frequency = 200 kHz.

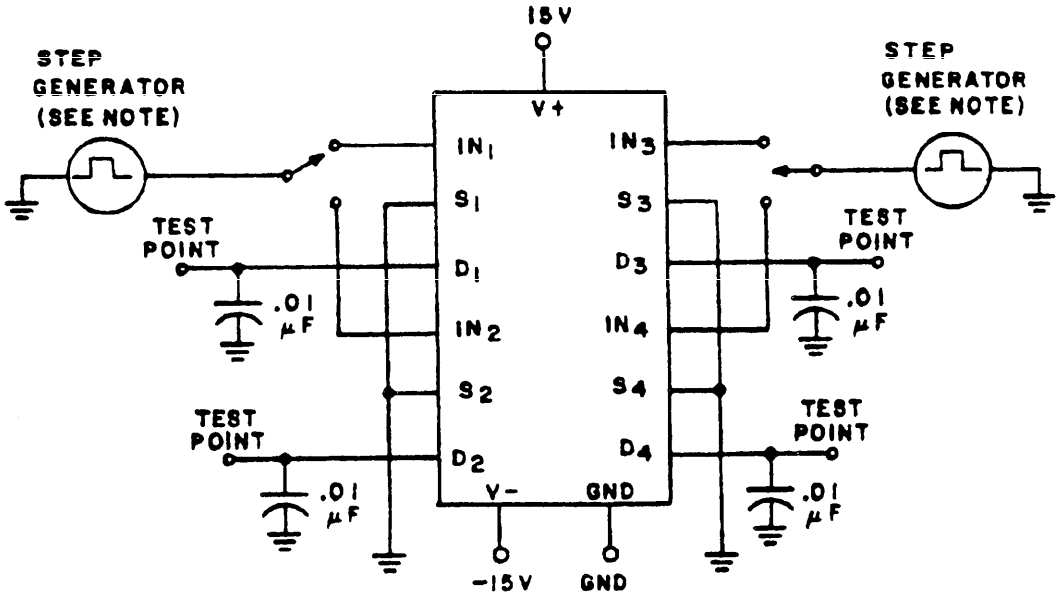
FIGURE 15. Crosstalk test circuits.

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Device type 01



Device type 02



NOTE: The pulse generator has the following characteristics: V_{GEN} = 0 to 4 V, rise time ≤20 ns, fall time ≤20 ns, PRR = 200 kHz.

FIGURE 16. Charge transfer error test circuits.

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TABLE III. Group A inspection for device type 01.

Subgroup	Symbol	MIL-STD-883 method	Case C	Terminal conditions 1/														Measured terminal	Test limits		Unit
				1	2	3	4	5	6	7	8	9	10	11	12	13	14				
			Test no.	IN_2	NC	GND	NC	S_2	U_2	V-	V_{REF}	U_1	S_1	NC	V^+	NC	IN_1		Min	Max	
1 T_A = 25°C	R_{DS}		1	0.8 V		GND				-15 V		-1 mA	10 V		15 V		0.8 V	9-10		70	Ω
			2	"		"				-15 V					15 V		"	6-5		70	"
			3	"		"		10 V	-1 mA	-10 V		-1 mA	7.5 V		10 V		"	9-10		100	"
			4	"		"		7.5 V	-1 mA	-10 V					10 V		"	6-5		100	"
			5	"		"				-15 V		1 mA	-10 V		15 V		"	9-10		70	"
			6	"		"		-10 V	1 mA	-15 V					15 V		"	6-5		70	"
			7	"		"				-10 V		1 mA	-7.5 V		10 V		"	9-10		100	"
			8	"		"		-7.5 V	1 mA	-10 V					10 V		"	6-5		100	"
	$I_{S(OFF)}$		9	2.4 V		"				-15 V		-14 V	14 V		15 V		2.4 V	10	-2	2	nA
			10	"		"		14 V	-14 V	"					"		"	5	"	"	"
			11	"		"				"		14 V	-14 V		"		"	10	"	"	"
			12	"		"		-14 V	14 V	"					"		"	5	"	"	"
	$I_{D(OFF)}$		13	"		"				"		-14 V	14 V		"		"	9	"	"	"
			14	"		"		14 V	-14 V	"					"		"	6	"	"	"
			15	"		"				"		14 V	-14 V		"		"	9	"	"	"
			16	"		"		-14 V	14 V	"					"		"	6	"	"	"
	$I_{D(ON)}$		17	0.8 V		"				"		14 V	14 V		"		0.8 V	9,10	"	"	"
			18	"		"		14 V	14 V	"					"		"	5,6	"	"	"
			19	"		"				"		-14 V	-14 V		"		"	9,10	"	"	"
			20	"		"		-14 V	-14 V	"					"		"	5,6	"	"	"
	I_{IL}	3009	21	2.4 V		"				"					"		"	14	-0.5	0.5	μA
	I_{IL}	3009	22	0.8 V		"				"					"		2.4 V	1	"	"	"
	I_{IH}	3010	23	0.8 V		"				"					"		2.4 V	14	"	"	"
		"	24	2.4 V		"				"					"		0.8 V	1	"	"	"
		"	25	0.8 V		"				"					"		15.0 V	14	"	"	"
		"	26	15.0 V		"				"					"		0.8 V	1	"	"	"
	$+I_{CC}$	3005	27	0.8 V		"				"					"		0.8 V	12		1.5	mA
	$+I_{CC}$	"	28	2.4 V		"				"					"		2.4 V	12		1.5	"
	$-I_{CC}$	"	29	0.8 V		"				"					"		0.8 V	7	-1.5		"
	$-I_{CC}$	"	30	2.4 V		"				"					"		2.4 V	7	-1.5		"

See footnotes at end of table.

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TABLE III. Group A inspection for device type 01 - Continued.

Subgroup	Symbol	MIL-STD-883 method	Case C Test no.	Terminal conditions 1/ 1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Test limits Min	Test limits Max	Unit
				IN_2	NC	GND	NC	S_2	D_2	$V-$	V_{REF}	D_1	S_1	NC	V^+	NC	IN_1				
2 T_A = 125°C	R_{DS}		31	0.8 V		GND				-15 V		-1 mA	10 V		15 V		0.8 V	9-10		100	Ω
			32	"		"		10 V	-1 mA	-15 V					15 V		"	6-5		100	"
			33	"		"				-10 V		-1 mA	7.5 V		10 V		"	9-10		150	"
			34	"		"		7.5 V	-1 mA	-10 V					10 V		"	6-5		150	"
			35	"		"				-15 V		1 mA	-10 V		15 V		"	9-10		100	"
			36	"		"		-10 V	1 mA	-15 V					15 V		"	6-5		100	"
			37	"		"				-10 V		1 mA	-7.5 V		10 V		"	9-10		150	"
			38	"		"		-7.5 V	1 mA	-10 V					10 V		"	6-5		150	"
	$I_{S(OFF)}$		39	2.4 V		"				-15 V		-14 V	14 V		15 V		2.4 V	10	-100	100	nA
			40	"		"		14 V	-14 V	"					"		"	5	"	"	"
			41	"		"				"		14 V	-14 V		"		"	10	"	"	"
			42	"		"		-14 V	14 V	"					"		"	5	"	"	"
	$I_{D(OFF)}$		43	"		"				"		-14 V	14 V		"		"	9	"	"	"
			44	"		"		14 V	-14 V	"					"		"	6	"	"	"
			45	"		"				"		14 V	-14 V		"		"	9	"	"	"
			46	"		"		-14 V	14 V	"					"		"	6	"	"	"
	$I_{D(ON)}$		47	0.8 V		"				"		14 V	14 V		"		0.8 V	9,10	-200	200	"
			48	"		"		14 V	14 V	"					"		"	5,6	"	"	"
			49	"		"				"		-14 V	-14 V		"		"	9,10	"	"	"
			50	"		"		-14 V	-14 V	"					"		"	5,6	"	"	"
	I_{IL}	3009	51	2.4 V		"				"					"		"	14	-1.0	1.0	μA
	I_{IL}	3009	52	0.8 V		"				"					"		2.4 V	1	"	"	"
	I_{IH}	3010	53	0.8 V		"				"					"		2.4 V	14	"	"	"
		"	54	2.4 V		"				"					"		0.8 V	1	"	"	"
		"	55	0.8 V		"				"					"		15.0 V	14	"	"	"
		"	56	15.0 V		"				"					"		0.8 V	1	"	"	"
	$+I_{CC}$	3005	57	0.8 V		"				"					"		0.8 V	12		1.5	mA
	$+I_{CC}$	"	58	2.4 V		"				"					"		2.4 V	12		1.5	"
	$-I_{CC}$	"	59	0.8 V		"				"					"		0.8 V	7	-1.5		"
	$-I_{CC}$	"	60	2.4 V		"				"					"		2.4 V	7	-1.5		"
3 T_A = -55°C	R_{DS}		61	0.8 V		"				"		-1 mA	10 V		"		0.8 V	9-10		70	Ω
			62	"		"		10 V	-1 mA	"					"		"	6-5		70	"
			63	"		"				-10 V		-1 mA	7.5 V		10 V		"	9-10		100	"
			64	"		"		7.5 V	-1 mA	-10 V					10 V		"	6-5		100	"
			65	"		"				-15 V		1 mA	-10 V		15 V		"	9-10		70	"
			66	"		"		-10 V	1 mA	-15 V					15 V		"	6-5		70	"
			67	"		"				-10 V		1 mA	-7.5 V		10 V		"	9-10		100	"
			68	"		"		-7.5 V	1 mA	-10 V					10 V		"	6-5		100	"
	$I_{S(OFF)}$		69	2.4 V		"				-15 V		-14 V	14 V		15 V		2.4 V	10	-100	100	nA
			70	"		"		14 V	-14 V	"					"		"	5	"	"	"
			71	"		"				"		14 V	-14 V		"		"	10	"	"	"
			72	"		"		-14 V	14 V	"					"		"	5	"	"	"

See footnotes at end of table.

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TABLE III. Group A inspection for device type 01 - Continued.

Subgroup	Symbol	MIL-STD-883 method	Case C Test no.	Terminal conditions 1/ 1 IN_2	2 NC	3 GND	4 NC	5 S_2	6 D_2	7 V-	8 V_{REF}	9 D_1	10 S_1	11 NC	12 V+	13 NC	14 IN_1	Measured terminal	Test limits Min	Test limits Max	Unit
3 T_A = -55°C	$I_{D(OFF)}$		73	2.4 V		GND				-15 V		-14 V	14 V		15 V		2.4 V	9	-100	100	nA
			74	"		"		14 V	-14 V	"					"		"	6	"	"	"
			75	"		"				"		14 V	-14 V		"		"	8	"	"	"
			76	"		"		-14 V	14 V	"					"		"	6	"	"	"
	$I_{D(ON)}$		77	0.8 V		"				"		14 V	14 V		"		0.8 V	9,10	-200	200	"
			78	"		"		14 V	14 V	"					"		"	5,6	"	"	"
			79	"		"				"		-14 V	-14 V		"		"	9,10	"	"	"
			80	"		"		-14 V	-14 V	"					"		"	5,6	"	"	"
	I_{IL}	3009	81	2.4 V		"				"					"		"	14	-1.0	1.0	µA
	I_{IL}	3009	82	0.8 V		"				"					"		2.4 V	1	"	"	"
	I_{IH}	3010	83	0.8 V		"				"					"		2.4 V	14	"	"	"
		"	84	2.4 V		"				"					"		0.8 V	1	"	"	"
		"	85	0.8 V		"				"					"		15.0 V	14	"	"	"
		"	86	15.0 V		"				"					"		0.8 V	1	"	"	"
	$+I_{CC}$	3005	87	0.8 V		"				"					"		0.8 V	12		2.0	mA
	$+I_{CC}$	"	88	2.4 V		"				"					"		2.4 V	12		2.0	"
	$-I_{CC}$	"	89	0.8 V		"				"					"		0.8 V	7	-2.0		"
	$-I_{CC}$	"	90	2.4 V		"				"					"		2.4 V	7	-2.0		"
4 T_A = 25°C	C_A 2/	3012	91			"				"					"		2/ 0 V	14		15	pF
	C_A 2/	"	92	2/ 0 V		"				"					"		"	1		"	"
	C_{IS} 2/	"	93			"				"			2/		"		5 V	10		"	"
	C_{IS} 2/	"	94	5 V		"		2/		"					"		"	5		"	"
	C_{OS} 3/	"	95			"				"		3/			"		5 V	9		20	"
	C_{OS} 3/	"	96	5 V		"			3/	"					"		"	6		20	"
9 T_A = 25°C	t_{ON}	3003	97			"				"		OUT	10 V		"		IN	14-9		600	ns
		fig. 13	98	IN		"		10 V	OUT	"					"		"	1-6		"	"
		"	99			"				"		OUT	-10 V		"		IN	14-9		"	"
		"	100	IN		"		-10 V	OUT	"					"		"	1-6		"	"
	t_{OFF}	"	101			"				"		OUT	10 V		"		IN	14-9		500	"
		"	102	IN		"		10 V	OUT	"					"		"	1-6		"	"
		"	103			"				"		OUT	-10 V		"		IN	14-9		"	"
		"	104	IN		"		-10 V	OUT	"					"		"	1-6		"	"

See footnotes at end of table.

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TABLE III. Group A inspection for device type 01 - Continued.

Subgroup	Symbol	MIL-STD-883 method	Case C Test no.	Terminal conditions 1/ 1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Test limits Min	Test limits Max	Unit
				IN_2	NC	GND	NC	S_2	D_2	V^-	V_{REF}	D_1	S_1	NC	V^+	NC	IN_1				
10 T_A = 125°C	t_{ON}	3003 Fig. 13	105 106 107 108	IN IN		GND		10 V -10 V	OUT OUT	-15 V		OUT OUT	10 V -10 V		15 V		IN IN	14-9 1-6 14-9 1-6		800	ns
	t_{OFF}		109 110 111 112	IN IN				10 V -10 V	OUT OUT			OUT OUT	10 V -10 V				IN IN	14-9 1-6 14-9 1-6		650	
11 T_A = -55°C	t_{ON}		113 114 115 116	IN IN				10 V -10 V	OUT OUT			OUT OUT	10 V -10 V				IN IN	14-9 1-6 14-9 1-6		600	
	t_{OFF}		117 118 119 120	IN IN				10 V -10 V	OUT OUT			OUT OUT	10 V -10 V				IN IN	14-9 1-6 14-9 1-6		500	
12 T_A = 25°C	V_{ISO} V_{ISO}	Fig. 14 Fig. 14	121 122	2.4 V 2.4 V				4/ IN					4/ IN				2.4 V 2.4 V	9 6	60		dB
13 T_A = 25°C	V_{CT}	Fig. 15	123	0.8 V				4/ IN									2.4 V	9			
14 T_A = 25°C	V_{CTE}	Fig. 16 Fig. 16	124 125	2.4 V 5/ IN				GND					GND				5/ IN 2.4 V	9 6	-10 -10	10 10	mV mV

See footnotes at end of table.

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TABLE III. Group A inspection for device type 01 - Continued.

Subgroup	Symbol	MIL-STD-883 method	Case I Test no.	Terminal conditions 1/ 1	2	3	4	5	6	7	8	9	10	Measured terminal	Test limits Min	Max	Unit
				IN_1	IN_2	GND	S_2	D_2	V-	V_{REF}	D_1	S_1	V+				
1 T_A = 25°C	R_{DS}		1	0.8 V	0.8 V	GND			-15 V		-1 mA	10 V	15 V	9-8		70	Ω
			2	"	"	"	10 V	-1 mA	-15 V				15 V	4-5		70	"
			3	"	"	"			-10 V		-1 mA	7.5 V	10 V	9-8		100	"
			4	"	"	"	7.5 V	-1 mA	-10 V				10 V	4-5		100	"
			5	"	"	"			-15 V		1 mA	-10 V	15 V	8-9		70	"
			6	"	"	"	-10 V	1 mA	-15 V				15 V	5-4		70	"
			7	"	"	"			-10 V		1 mA	-7.5 V	10 V	8-9		100	"
			8	"	"	"	-7.5 V	1 mA	-10 V				10 V	5-4		100	"
	$I_{S(OFF)}$		9	2.4 V	2.4 V	"			-15 V		-14 V	14 V	15 V	9	-2	2	nA
			10	"	"	"	14 V	-14 V	"				"	4	"	"	"
			11	"	"	"			"		14 V	-14 V	"	9	"	"	"
			12	"	"	"	-14 V	14 V	"				"	4	"	"	"
	$I_{D(OFF)}$		13	"	"	"			"		14 V	-14 V	"	8	"	"	"
			14	"	"	"	-14 V	14 V	"				"	5	"	"	"
			15	"	"	"			"		-14 V	14 V	"	8	"	"	"
			16	"	"	"	14 V	-14 V	"				"	5	"	"	"
	$I_{D(ON)}$		17	0.8 V	0.8 V	"			"		14 V	14 V	"	8	"	"	"
			18	"	"	"	14 V	14 V	"				"	5	"	"	"
			19	"	"	"			"		-14 V	-14 V	"	8	"	"	"
			20	"	"	"	-14 V	-14 V	"				"	5	"	"	"
	I_{IL}	3009	21	"	2.4 V	"			"				"	1	-0.5	0.5	μA
	I_{IL}	3009	22	2.4 V	0.8 V	"			"				"	2	"	"	"
	I_{IN}	3010	23	2.4 V	0.8 V	"			"				"	1	"	"	"
		"	24	0.8 V	2.4 V	"			"				"	2	"	"	"
		"	25	15.0 V	0.8 V	"			"				"	1	"	"	"
		"	26	0.8 V	15.0 V	"			"				"	2	"	"	"
	$+I_{CC}$	3005	27	0.8 V	0.8 V	"			"				"	10		1.5	mA
	$+I_{CC}$	"	28	2.4 V	2.4 V	"			"				"	10		1.5	"
	$-I_{CC}$	"	29	0.8 V	0.8 V	"			"				"	6	-1.5		"
	$-I_{CC}$	"	30	2.4 V	2.4 V	"			"				"	6	-1.5		"
2 T_A = 125°C	R_{DS}		31	0.8 V	0.8 V	"			"		-1 mA	10 V	"	9-8		100	Ω
			32	"	"	"	10 V	-1 mA	"				"	4-5		100	"
			33	"	"	"			-10 V		-1 mA	7.5 V	10 V	9-8		150	"
			34	"	"	"	7.5 V	-1 mA	-10 V				10 V	4-5		150	"
			35	"	"	"			-15 V		1 mA	-10 V	15 V	9-8		100	"
			36	"	"	"	-10 V	1 mA	-15 V				15 V	4-5		100	"
			37	"	"	"			-10 V		1 mA	-7.5 V	10 V	9-8		150	"
			38	"	"	"	-7.5 V	1 mA	-10 V				10 V	4-5		150	"
	$I_{S(OFF)}$		39	2.4 V	2.4 V	"			-15 V		-14 V	14 V	15 V	9	-100	100	nA
			40	"	"	"	14 V	-14 V	"				"	4	"	"	"
			41	"	"	"			"		14 V	-14 V	"	9	"	"	"
			42	"	"	"	-14 V	14 V	"				"	4	"	"	"
	$I_{D(OFF)}$		43	"	"	"			"		-14 V	14 V	"	8	"	"	"
			44	"	"	"	14 V	-14 V	"				"	5	"	"	"
			45	"	"	"			"		14 V	-14 V	"	8	"	"	"
			46	"	"	"	-14 V	14 V	"				"	5	"	"	"

See footnotes at end of table.

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TABLE III. Group A inspection for device type 01 - Continued.

Subgroup	Symbol	MIL-STD-883 method	Case I test no.	Terminal conditions 1/ 1 IN_1	2 IN_2	3 GND	4 S_2	5 D_2	6 V_-	7 V_{REF}	8 D_1	9 S_1	10 V_+	Measured terminal	Test limits Min	Max	Unit
2 T_A = 125°C	$I_{D(ON)}$		47	0.8 V	0.8 V	GND	14 V	14 V	-15 V		14 V	14 V	15 V	8	-200	200	nA
			48	"	"	"			"				"	5	"	"	"
			49	"	"	"	-14 V	-14 V	"		-14 V	-14 V	"	8	"	"	"
			50	"	"	"			"				"	5	"	"	"
	I_{IL}	3009	51	"	2.4 V	"			"				"	1	-1.0	1.0	µA
	I_{IL}	3009	52	2.4 V	0.8 V	"			"				"	2	"	"	"
	I_{IH}	3010	53	2.4 V	0.8 V	"			"				"	1	"	"	"
		"	54	0.8 V	2.4 V	"			"				"	2	"	"	"
		"	55	15.0 V	0.8 V	"			"				"	1	"	"	"
		"	56	0.8 V	15.0 V	"			"				"	2	"	"	"
	$+I_{CC}$	3005	57	0.8 V	0.8 V	"			"				"	10		1.5	mA
	$+I_{CC}$	"	58	2.4 V	2.4 V	"			"				"	10		1.5	"
	$-I_{CC}$	"	59	0.8 V	0.8 V	"			"				"	6	-1.5		"
	$-I_{CC}$	"	60	2.4 V	2.4 V	"			"				"	6	-1.5		"
3 T_A = -55°C	R_{DS}		61	0.8 V	0.8 V	"	10 V	-1 mA	"		-1 mA	10 V	"	9-8		70	Ω
			62	"	"	"			-10 V				"	6-5		70	"
			63	"	"	"	7.5 V	-1 mA	-10 V		-1 mA	7.5 V	10 V	9-8		100	"
			64	"	"	"			-15 V				10 V	6-5		100	"
			65	"	"	"	-10 V	1 mA	-15 V		1 mA	-10 V	15 V	9-8		70	"
			66	"	"	"			-10 V				15 V	6-5		70	"
			67	"	"	"	-7.5 V	1 mA	-10 V		1 mA	-7.5 V	10 V	9-8		100	"
			68	"	"	"			-10 V				10 V	6-5		100	"
	$I_{S(OFF)}$		69	2.4 V	2.4 V	"	14 V	-14 V	-15 V		-14 V	14 V	15 V	9	-100	100	nA
			70	"	"	"			"				"	4	"	"	"
			71	"	"	"	-14 V	14 V	"		14 V	-14 V	"	9	"	"	"
			72	"	"	"			"				"	4	"	"	"
	$I_{D(OFF)}$		73	"	"	"	14 V	-14 V	"		-14 V	14 V	"	8	"	"	"
			74	"	"	"			"				"	5	"	"	"
			75	"	"	"	-14 V	14 V	"		14 V	-14 V	"	8	"	"	"
			76	"	"	"			"				"	5	"	"	"
	$I_{D(ON)}$		77	0.8 V	0.8 V	"	14 V	14 V	"		14 V	14 V	"	8	-200	200	"
			78	"	"	"			"				"	5	"	"	"
			79	"	"	"	-14 V	-14 V	"		-14 V	-14 V	"	8	"	"	"
			80	"	"	"			"				"	5	"	"	"
	I_{IL}	3009	81	"	2.4 V	"			"				"	1	-1.0	1.0	µA
	I_{IL}	3009	82	2.4 V	0.8 V	"			"				"	2	"	"	"
	I_{IH}	3010	83	2.4 V	0.8 V	"			"				"	1	"	"	"
		"	84	0.8 V	2.4 V	"			"				"	2	"	"	"
		"	85	15.0 V	0.8 V	"			"				"	1	"	"	"
		"	86	0.8 V	15.0 V	"			"				"	2	"	"	"
	$+I_{CC}$	3005	87	0.8 V	0.8 V	"			"				"	10		2.0	mA
	$+I_{CC}$	"	88	2.4 V	2.4 V	"			"				"	10		2.0	"
	$-I_{CC}$	"	89	0.8 V	0.8 V	"			"				"	6	-2.0		"
	$-I_{CC}$	"	90	2.4 V	2.4 V	"			"				"	6	-2.0		"

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TABLE III. Group A inspection for device type 01 - Continued.

Subgroup	Symbol	MIL-STD-883 method	Case T Test no.	Terminal conditions 1/ 1	2	3	4	5	6	7	8	9	10	Measured terminal	Test limits Min	Max	Unit
				IN_1	IN_2	GND	S_2	D_2	V-	V_{REF}	D_1	S_1	V+				
4 T_A = 25°C	C_A 2/ C_A	3012 "	91 92	2/ 0 V	2/ 0 V	GND "			-15 V "				15 V "	1 2		15 "	pF "
	C_{IS} 2/ C_{IS}	" "	93 94	5 V	5 V	" "	2/		" "			2/	" "	9 4		" "	" "
	C_{OS} 3/ C_{OS} 3/	" "	95 96	5 V	5 V	" "		3/	" "		3/		" "	0 5		20 20	" "
9 T_A = 25°C	t_{ON}	3003 Fig. 13 " "	97 98 99 100	IN IN	IN IN	" " " "	10 V -10 V	OUT OUT	" " " "		OUT OUT	10 V -10 V	" " " "	1-0 2-5 1-0 2-5		600 " " "	ns " " "
	t_{OFF}	" " " "	101 102 103 104	IN IN	IN IN	" " " "	10 V -10 V	OUT OUT	" " " "		OUT OUT	10 V -10 V	" " " "	1-0 2-5 1-0 2-5		500 " " "	" " " "
10 T_A = 125°C	t_{ON}	" " " "	105 106 107 108	IN IN	IN IN	" " " "	10 V -10 V	OUT OUT	" " " "		OUT OUT	10 V -10 V	" " " "	1-0 2-5 1-0 2-5		800 " " "	" " " "
	t_{OFF}	" " " "	109 110 111 112	IN IN	IN IN	" " " "	10 V -10 V	OUT OUT	" " " "		OUT OUT	10 V -10 V	" " " "	1-0 2-5 1-0 2-5		650 " " "	" " " "
11 T_A = -55°C	t_{ON}	" " " "	113 114 115 116	IN IN	IN IN	" " " "	10 V -10 V	OUT OUT	" " " "		OUT OUT	10 V -10 V	" " " "	1-0 2-5 1-0 2-5		600 " " "	" " " "
	t_{OFF}	" " " "	117 118 119 120	IN IN	IN IN	" " " "	10 V -10 V	OUT OUT	" " " "		OUT OUT	10 V -10 V	" " " "	1-0 2-5 1-0 2-5		500 " " "	" " " "
12 T_A = 25°C	V_{ISO} V_{ISO}	Fig. 14 Fig. 14	121 122	2.4 V "	2.4 V "	" "	IN		" "			IN	" "	8 5	60 "		dB "
13 T_A = 25°C	V_{CT}	Fig. 15	123	"	"	"			"				"	8	"		"
14 T_A = 25°C	V_{CTE}	Fig. 16	124 125	IN 2.4 V	" IN	" "	GND		" "			GND	" "	5 8	-10 -10	10 10	mV mV

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TABLE III. Group A inspection for device type 02.

Subgroup	Symbol	MIL-STD-883 method	Case E Test no.	Terminal conditions 1/ 1 IN_1	2 D_1	3 S_1	4 V-	5 GND	6 S_4	7 D_4	8 IN_4	9 IN_3	10 D_3	11 S_3	12 V_{REF}	13 V+	14 S_2	15 D_2	16 IN_2	Measured terminal	Test limits Min	Test limits Max	Unit
1 T_A = 25°C	R_{DS}		1	0.8 V	-1 mA	10 V	-15 V	GND			0.8 V	0.8 V				15 V			0.8 V	3-2		175	Ω
			2	"			"	"			"	"				"	10 V	-1 mA	"	14-15		"	"
			3	"			"	"			"	"	-1 mA	10 V		"			"	11-10		"	"
			4	"			"	"	10 V	-1 mA	"	"				"			"	6-7		"	"
			5	"	-1 mA	7.5 V	-10 V	"			"	"				10 V			"	3-2		200	"
			6	"			"	"			"	"				"	7.5 V	-1 mA	"	14-15		"	"
			7	"			"	"			"	"	-1 mA	7.5 V		"			"	11-10		"	"
			8	"			"	"	7.5 V	-1 mA	"	"				"			"	6-7		"	"
			9	"	1 mA	-10 V	-15 V	"			"	"				15 V			"	2-3		175	"
			10	"			"	"			"	"				"	-10 V	1 mA	"	15-14		"	"
			11	"			"	"			"	"	1 mA	-10 V		"			"	10-11		"	"
			12	"			"	"	-10 V	1 mA	"	"				"			"	7-6		"	"
			13	"	1 mA	-7.5 V	-10 V	"			"	"				10 V			"	2-3		200	"
			14	"			"	"			"	"				"	-7.5 V	1 mA	"	15-14		"	"
			15	"			"	"			"	"	1 mA	-7.5 V		"			"	10-11		"	"
			16	"			"	"	-7.5 V	1 mA	"	"				"			"	7-6		"	"
	$I_{S(OFF)}$		17	2.4 V	-14 V	14 V	-15 V	"			2.4 V	2.4 V				15 V			2.4 V	3	-2	2	nA
			18	"			"	"			"	"				"	14 V	-14 V	"	14	"	"	"
			19	"			"	"			"	"	-14 V	14 V		"			"	11	"	"	"
			20	"			"	"	14 V	-14 V	"	"				"			"	6	"	"	"
			21	"	14 V	-14 V	"	"			"	"				"	-14 V	14 V	"	3	"	"	"
			22	"			"	"			"	"				"			"	14	"	"	"
			23	"			"	"			"	"	14 V	-14 V		"			"	11	"	"	"
			24	"			"	"	-14 V	14 V	"	"				"			"	6	"	"	"
	$I_{D(OFF)}$		25	"	-14 V	14 V	"	"			"	"				"			"	2	"	"	"
			26	"			"	"			"	"				"	14 V	-14 V	"	15	"	"	"
			27	"			"	"			"	"	-14 V	14 V		"			"	10	"	"	"
			28	"			"	"	14 V	-14 V	"	"				"			"	7	"	"	"
			29	"	14 V	-14 V	"	"			"	"				"			"	2	"	"	"
			30	"			"	"			"	"				"	-14 V	14 V	"	15	"	"	"
			31	"			"	"			"	"	14 V	-14 V		"			"	10	"	"	"
			32	"			"	"	-14 V	14 V	"	"				"			"	7	"	"	"
	$I_{D(ON)}$		33	0.8 V	14 V	14 V	"	"			0.8 V	0.8 V				"			0.8 V	2	"	"	"
			34	"			"	"			"	"				"	14 V	14 V	"	15	"	"	"
			35	"			"	"			"	"	14 V	14 V		"			"	10	"	"	"
			36	"			"	"	14 V	14 V	"	"				"			"	7	"	"	"
			37	"	-14 V	-14 V	"	"			"	"				"			"	2	"	"	"
			38	"			"	"			"	"				"	-14 V	-14 V	"	15	"	"	"
			39	"			"	"			"	"	-14 V	-14 V		"			"	10	"	"	"
			40	"			"	"	-14 V	-14 V	"	"				"			"	7	"	"	"
	I_{IL}	3009	41	"			"	"			2.4 V	2.4 V				"			2.4 V	1	-0.5	0.5	μA
		"	42	2.4 V			"	"			"	2.4 V				"			0.8 V	16	"	"	"
		"	43	"			"	"			"	0.8 V				"			2.4 V	9	"	"	"
		"	44	"			"	"			0.8 V	2.4 V				"			2.4 V	8	"	"	"

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TABLE III. Group A inspection for device type 02 - Continued.

Subgroup	Symbol	MIL-STD-883 method	Case E test no.	Terminal conditions 1/ 1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Test limits Min	Test limits Max	Unit
				IN_1	O_1	S_1	$V-$	GND	S_4	O_4	IN_4	IN_3	O_3	S_3	V_{REF}	$V+$	S_2	O_2	IN_2				
1 T_A = 25°C	I_{IH}	3010	45	2.4 V			-15 V	Gnd			0.8 V	0.8 V				15 V			0.8 V	1	-0.5	0.5	mA
			46	0.8 V								0.8 V							2.4 V	16			
			47									2.4 V							0.8 V	9			
			48								2.4 V	0.8 V							0.8 V	8			
			49	15.0 V							0.8 V	0.8 V							0.8 V	1			
			50	0.8 V								0.8 V							15.0 V	16			
			51									15.0 V							0.8 V	9			
			52								15.0 V	0.8 V							0.8 V	8			
	$+I_{CC}$	3005	53								0.8 V	0.8 V							0.8 V	13		1.5	mA
	$+I_{CC}$		54	2.4 V							2.4 V	2.4 V							2.4 V	13		1.5	
	$-I_{CC}$		55	0.8 V							0.8 V	0.8 V							0.8 V	4	-1.5		
	$-I_{CC}$		56	2.4 V							2.4 V	2.4 V							2.4 V	4	-1.5		
2 T_A = 125°C	R_{DS}		57	0.8 V	-1 mA	10 V					0.8 V	0.8 V							0.8 V	3-2		250	Ω
			58														10 V	-1 mA		14-15			
			59										-1 mA	10 V						11-10			
			60						10 V	-1 mA										6-7			
			61		-1 mA	7.5 V	-10 V									10 V				3-2			
			62														7.5 V	-1 mA		14-15			
			63										-1 mA	7.5 V						11-10			
			64						7.5 V	-1 mA										6-7			
			65		1 mA	-10 V	-15 V									15 V				2-3			
			66														-10 V	1 mA		15-14			
			67										1 mA	-10 V						10-11			
			68						-10 V	1 mA										7-6			
			69		1 mA	-7.5 V	-10 V									10 V				2-3			
			70														-7.5 V	1 mA		15-14			
			71										1 mA	-7.5 V						10-11			
			72						-7.5 V	1 mA										7-6			
	$I_{S(OFF)}$		73	2.4 V	-14 V	14 V	-15 V				2.4 V	2.4 V				15 V			2.4 V	3	-100		nA
			74														14 V	-14 V		14			
			75										-14 V	14 V						11			
			76						14 V	-14 V										6			
			77		14 V	-14 V														3			
			78														-14 V	14 V		14			
			79										14 V	-14 V						11			
			80						-14 V	14 V										6			
	$I_{D(OFF)}$		81		-14 V	14 V														2			
			82														14 V	-14 V		15			
			83										-14 V	14 V						10			
			84						14 V	-14 V										7			
			85	0.8 V	14 V	-14 V					0.8 V	0.8 V							0.8 V	2			
			86														-14 V	14 V		15			
			87										14 V	-14 V						10			
			88						-14 V	14 V										7			
	$I_{D(ON)}$		89		14 V	14 V														2	-200	200	
			90														14 V	14 V		15			
			91										14 V	14 V						10			
			92						14 V	14 V										7			
			93		-14 V	-14 V														2			
			94														-14 V	-14 V		15			
			95										-14 V	-14 V						10			
			96						-14 V	-14 V										7			

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TABLE III. Group A inspection for device type 02 - Continued.

Subgroup	Symbol	MIL-STD-883 method	Case E Test no.	Terminal conditions 1/ 1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Test limits Min	Max	Unit
				IN_1	O_1	S_1	V-	GND	S_4	O_4	IN_4	IN_3	O_3	S_3	V_{REF}	V+	S_2	O_2	IN_2				
2 T_A = 125°C	I_{IL}	3009	97	0.8 V			-15 V	GND			2.4 V	2.4 V				15 V			2.4 V	1	-1.0	1.0	µA
			98	2.4 V								2.4 V							0.8 V	16			
			99									0.8 V							2.4 V	9			
			100								0.8 V	2.4 V							2.4 V	8			
	I_{IH}	3010	101									0.8 V							0.8 V	1			
			102	0.8 V								0.8 V							2.4 V	16			
			103	0.8 V								2.4 V								9			
			104	0.8 V							2.4 V	0.8 V								8			
			105	15.0 V							0.8 V								0.8 V	1			
			106	0.8 V															15.0 V	16			
			107									15.0 V							0.8 V	9			
			108								15.0 V	0.8 V								8			

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TABLE III. Group A inspection for device type 02 - Continued.

Subgroup	Symbol	MIL-STD-883 method	Case E Test no.	Terminal conditions 1/ 1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Test limits Min	Test limits Max	Unit
				IN_1	D_1	S_1	V^-	GND	S_4	D_4	IN_4	IN_3	D_3	S_3	V_{REF}	V^+	S_2	D_2	IN_2				
3 T_A = -55°C	$I_{D(ON)}$	3009	145	0.8 V	14 V	14 V	-15 V	GND			0.8 V	0.8 V				15 V			0.8 V	2	-200	200	nA
			146	"			"	"			"	"				"	14 V	14 V	"	15	"	"	
			147	"			"	"			"	"	14 V	14 V		"			"	10	"	"	
			148	"			"	"	14 V	14 V	"	"				"			"	7	"	"	
			149	"	-14 V	-14 V	"	"			"	"				"			"	2	"	"	
			150	"			"	"			"	"				"	-14 V	-14 V	"	15	"	"	
			151	"			"	"			"	"	-14 V	-14 V		"			"	10	"	"	
			152	"			"	"	-14 V	-14 V	"	"				"			"	7	"	"	
	I_{IL}	"	153	"			"	"			2.4 V	2.4 V				"			2.4 V	1	-1.0	1.0	mA
		"	154	2.4 V			"	"			"	2.4 V				"			0.8 V	16	"	"	
		"	155	"			"	"			"	0.8 V				"			2.4 V	9	"	"	
		"	156	"			"	"			0.8 V	2.4 V				"			2.4 V	8	"	"	
	I_{IH}	3010	157	"			"	"			"	0.8 V				"			0.8 V	1	"	"	"
		"	158	0.8 V			"	"			"	0.8 V				"			2.4 V	16	"	"	
		"	159	"			"	"			"	2.4 V				"			0.8 V	9	"	"	
		"	160	"			"	"			2.4 V	0.8 V				"			"	8	"	"	
		"	161	15.0 V			"	"			0.8 V	"				"			"	1	"	"	
		"	162	0.8 V			"	"			"	"				"			15.0 V	16	"	"	
		"	163	"			"	"			"	15.0 V				"			0.8 V	9	"	"	
		"	164	"			"	"			15.0 V	0.8 V				"			0.8 V	8	"	"	
	$+I_{CC}$	3005	165	"			"	"			0.8 V	0.8 V				"			0.8 V	13		2.0	mA
	$+I_{CC}$	"	166	2.4 V			"	"			2.4 V	2.4 V				"			2.4 V	13		2.0	
	$-I_{CC}$	"	167	0.8 V			"	"			0.8 V	0.8 V				"			0.8 V	4	-2.0		"
	$-I_{CC}$	"	168	2.4 V			"	"			2.4 V	2.4 V				"			2.4 V	4	-2.0		
4 T_A = 25°C	C_A 2/	3012	169	2/ 0 V			"	"								"				1		15	pF
		"	170	"			"	"				2/ 0 V				"			2/ 0 V	16		"	
		"	171	"			"	"				"				"			"	9		"	
		"	172	"			"	"			2/ 0 V	"				"			"	8		"	
	C_{IS} 2/	"	173	2/ 5 V			"	"								"				3			"
		"	174	"			"	"				2/ 5 V				"			2/ 5 V	14			
		"	175	"			"	"				"				"			"	11			
		"	176	"			"	"			2/ 5 V	"				"			"	6			
	C_{OS} 3/	"	177	3/ 5 V			"	"								"				2		20	"
		"	178	"			"	"				3/ 5 V				"			3/ 5 V	15		"	
		"	179	"			"	"				"				"			"	10		"	
		"	180	"			"	"			3/ 5 V	"				"			"	7		"	
9 T_A = 25°C	t_{ON}	3003 Fig. 13	181	IN	OUT	10 V	"	"								"				1-2		600	ns
		"	182	"	"	"	"	"				IN	OUT	10 V		"	10 V	OUT	IN	16-15		"	
		"	183	"	"	"	"	"				"	"	"		"	"	"	"	9-10		"	
		"	184	"	"	"	"	"	10 V	OUT	IN	"	"	"		"	"	"	"	8-7		"	
		"	185	IN	OUT	-10 V	"	"								"				1-2		"	
		"	186	"	"	"	"	"				IN	OUT	-10 V		"	-10 V	OUT	IN	16-15		"	
		"	187	"	"	"	"	"				"	"	"		"	"	"	"	9-10		"	
		"	188	"	"	"	"	"	-10 V	OUT	IN	"	"	"		"	"	"	"	8-7		"	

See footnotes at end of table.

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TABLE III. Group A inspection for device type 02 - Continued.

Subgroup	Symbol	MIL-STD-883 method	Case E Test no.	Terminal conditions 1/ 1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Test limits Min	Max	Unit
				IN_1	O_1	S_1	V^-	GND	S_4	O_4	IN_4	IN_3	O_3	S_3	V_{BIAS} 2/	V^+	S_2	O_2	IN_2				
9 T_A = 25°C	t_{OFF}	3003 Fig. 13	189	IN	OUT	10 V	-15 V	GND								15 V				1-8		500	ns
			190														10 V	OUT	IN	16-15			
			191									IN	OUT	10 V						9-10			
			192						10 V	OUT	IN									8-7			
			193	IN	OUT	-10 V														1-8			
			194														-10 V	OUT	IN	16-15			
			195									IN	OUT	-10 V						9-10			
			196						-10 V	OUT	IN									8-7			
10 T_A = 125°C	t_{ON}		197	IN	OUT	10 V														1-8		800	
			198														10 V	OUT	IN	16-15			
			199									IN	OUT	10 V						9-10			
			200						10 V	OUT	IN									8-7			
			201	IN	OUT	-10 V														1-8			
			202														-10 V	OUT	IN	16-15			
			203									IN	OUT	-10 V						9-10			
			204						-10 V	OUT	IN									8-7			
	t_{OFF}		205	IN	OUT	10 V														1-8		650	
			206														10 V	OUT	IN	16-15			
			207									IN	OUT	10 V						9-10			
			208						10 V	OUT	IN									8-7			
			209	IN	OUT	-10 V														1-8			
			210														-10 V	OUT	IN	16-15			
			211									IN	OUT	-10 V						9-10			
			212						-10 V	OUT	IN									8-7			
11 T_A = -55°C	t_{ON}		213	IN	OUT	10 V														1-8		600	
			214														10 V	OUT	IN	16-15			
			215									IN	OUT	10 V						9-10			
			216						10 V	OUT	IN									8-7			
			217	IN	OUT	-10 V														1-8			
			218														-10 V	OUT	IN	16-15			
			219									IN	OUT	-10 V						9-10			
			220						-10 V	OUT	IN									8-7			
	t_{OFF}		221	IN	OUT	10 V														1-8		500	
			222														10 V	OUT	IN	16-15			
			223									IN	OUT	10 V						9-10			
			224						10 V	OUT	IN									8-7			
			225	IN	OUT	-10 V														1-8			
			226														-10 V	OUT	IN	16-15			
			227									IN	OUT	-10 V						9-10			
			228						-10 V	OUT	IN									8-7			
12 T_A = 25°C	V_{ISO}	Fig. 14	229	2.4 V		IN					2.4 V	2.4 V						IN	2.4 V	2	60		dB
			230																	16			
			231											IN						10			
			232						IN											7			
13 T_A = 25°C	V_{CT}	Fig. 15	233	2.4 V							0.8 V	0.8 V		IN			IN		0.8 V	2			
			234	0.8 V		IN						0.8 V		IN					2.4 V	15			
			235									2.4 V		IN			IN		0.8 V	10			
			236								2.4 V	0.8 V		IN			IN		0.8 V	7			
14 T_A = 25°C	V_{CTE}	Fig. 16	237	IN		GND						2.4 V							2.4 V	2	-10	10	mV
			238	2.4 V								2.4 V					GND		IN	16			
			239									IN		GND					2.4 V	10			
			240						GND		IN	2.4 V							2.4 V	7			

1/ Pins not designated may be "high" level logic, "low" level logic, or open.
2/ Connect capacitance bridge between measured input terminal and GND, f = 1 MHz; see 4.4.1c.
3/ Connect capacitance bridge between measured output terminal and GND, f = 1 MHz; see 4.4.1c.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007, as applicable, of MIL-STD-883, except as modified herein.

4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

a. Burn-in tests (method 1015 of MIL-STD-883).

(1) Class S devices: Dynamic tests (test condition D or E) using the circuit shown on figure 4. Test duration shall be 240 hours minimum. Static tests (test condition A) using the circuit shown on figure 5. Test duration shall be 48 hours minimum.

(2) Class B devices: Test condition A, using the circuit shown on figure 5; or test condition D, using the circuit shown on figure 4; or test condition F, using the circuit shown on figure 6.

b. Interim and final electrical test parameters shall be as specified in table II herein.

c. The percent defective allowable (PDA) for class S devices shall be as specified in MIL-M-38510. The PDA for class B devices shall be 10 percent based on failures from group A, subgroup 1 test after cooldown as final electrical test in accordance with method 5004 of MIL-STD-883, and with no intervening electrical measurements. If interim electrical parameter tests are performed prior to burn-in, failures resulting from pre-burn-in screening may be excluded from the PDA. If interim electrical parameter tests prior to burn-in are omitted, then all screening failures shall be included in the PDA. The verified failures of group A, subgroup 1 after burn-in divided by the total number of devices submitted for burn-in in that lot shall be used to determine the percent defective for that lot, and the lot shall be accepted or rejected based on the PDA for the applicable device class.

4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510, and as specified herein. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Tests shall be as specified in table II herein.

b. Subgroups 5, 6, 7, and 8 shall be omitted.

c. Subgroup 4 (C_A, C_{IS}, and C_{OS} measurements) shall be measured only for initial qualification and after process or design changes which may affect input capacitance. Capacitance shall be measured between the designated terminal and ground at a frequency of 1 MHz.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883 and as follows:

a. Electrical parameters shall be as specified in table II herein. For class S devices, delta limits shall apply only to subgroup 5 of group B inspection.

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b. Steady state life test for class S devices shall be in accordance with table IIa of method 5005 of MIL-STD-883, using the circuit shown on figure 6. If the alternate burn-in conditions are used, the circuit shown on figure 4 shall be used.

c. A special subgroup shall be added using an LTPD of 15 for class S, B, and C devices. This subgroup shall consist of a high voltage test of the input protection circuits, V_{ZAP} (see 4.5.3).

4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II herein. Delta limits shall apply only to subgroup 1 of group C inspection for class B and C devices.

b. Steady state life test (method 1005 of MIL-STD-883) conditions:

(1) Test condition C, using the circuit shown on figure 5; or test condition D, using the circuit shown on figure 4; or test condition F, using the circuit shown on figure 6.

(2) T_A = 125°C minimum.

(3) Test duration: 1,000 hours, except as permitted by appendix B of MIL-M-38510 and method 1005 of MIL-STD-883.

c. A special subgroup shall be added to the group C inspection requirements for class B and C devices, and shall consist of the tests, conditions, and limits as specified for subgroups 10 and 11 of group A.

4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II herein.

b. Special subgroups shall be added using an LTPD of 15 for all classes. These subgroups shall consist of: Subgroup 12, off isolation test, subgroup 13, crosstalk test, and subgroup 14, charge transfer error test. These subgroups shall be measured only for initial qualification, and thereafter for group D tests at 25°C only.

TABLE IV. Delta limits at 25°C. 1/

Parameter 2/	Device types	
	01	02
R_{DS}	15Ω	15Ω
$I_{D(ON)}$	1 nA	1 nA

1/ Each test applies to every switch in the package.
2/ Each of the above parameters shall be recorded before and after the required burn-in and life tests to determine deltas (Δ).

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as specified herein.

4.5.1 Voltage and current. All voltages given are referenced to the microcircuit ground terminal, unless otherwise specified. Currents given are conventional and positive when flowing into the referenced terminal.

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4.5.2 Life test cooldown procedure. When devices are measured at 25°C following application of the operating life or burn-in test condition, they shall be cooled to room temperature prior to removal of the bias.

4.5.3 High voltage (V_{ZAP}) test of input protection circuits. All input terminals (up to a maximum of four) of the DUT shall be subjected to a voltage pulse from a 100 pF source charged to 400 V. This destructive test shall be conducted as follows using the test circuit on figure 17.

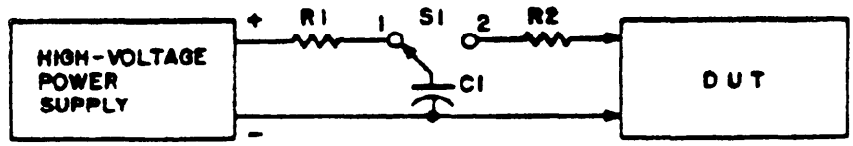
a. Measure I_{IL} and I_{IH} at the input selected, as stated above, at 25°C. The test limit for each input tested shall meet specified limits as cited in table I at $V_{(supply)} = \pm 15$ V.

b. V_{ZAP} is applied to the DUT in the following modes (see table V) by charging C_1 to V_{ZAP} with S1 in position 1 and then switching to position 2.

TABLE V. Modes for high-voltage test.

Mode	+ Terminal	- Terminal
1	V+	Address input
2	Address input	GND

c. Within 24 hours repeat the I_{IH}, and I_{IL} measurements on the same terminals as performed above. If the DUT exhibits leakage currents in excess of the specified limits after the V_{ZAP} test, it shall be classified as a failure.



V_{ZAP} = 400 V charge on C1.
10 MΩ ≤ R1 ≤ 50 MΩ.
R2 = 1.5 kΩ.
C1 = 100 pF.
S1 = Hg-wetted "bounceless" relay.

FIGURE 17. High voltage (V_{ZAP}) test circuit.

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4.6 Data reporting. When specified in the acquisition document, a copy of the following data, as applicable, shall be supplied:

a. Attributes data for all screening tests (see 4.2) and variables data for all static burn-in, dynamic burn-in, and operating life tests.

b. A copy of each radiograph.

c. The quality conformance inspection data (see 4.4).

d. Parameter distribution data on parameters evaluated during burn-in (see 3.5).

e. Final electrical parameter data (see 4.2b).

4.7 Inspection of packaging. Inspection of packaging shall be in accordance with MIL-M-38510.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510.

6. NOTES

6.1 Notes. The notes specified in MIL-M-38510 are applicable to this specification.

6.2 Intended use. Microcircuits conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.3 Ordering data. The acquisition document should specify the following:

a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

c. Requirements for certificate of compliance, if applicable.

d. Requirements for notification of change of product or process to acquiring activity in addition to notification to the qualifying activity, if applicable.

e. Requirements for packaging and packing.

f. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action, and reporting of results, if applicable.

g. Requirements for product assurance options.

h. Requirements for special carriers, lead lengths, or lead forming, if applicable. These requirements shall not affect the part number. Unless otherwise specified, these requirements shall not apply to direct purchase by or direct shipment to the Government.

i. Requirements for "JAN" marking.

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6.4 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331, and as follows:

V^+	Positive supply voltage.
V^-	Negative supply voltage.
V_{REF}	Reference voltage.
V_{IL}	Input (address) low voltage.
V_{IH}	Input (address) high voltage.
I_{CC+}	Positive supply current.
I_{CC-}	Negative supply current.
$I_{S(OFF)}$	Leakage current into the source terminal of an "OFF" switch.
$I_{D(OFF)}$	Leakage current into the drain terminal of an "OFF" switch.
$I_{D(ON)}$	Leakage current from an "ON" driver into the switch.
I_{IL}	Low level (address) input current.
I_{IH}	High level (address) input current.
R_{DS}	Resistance of an "ON" switch.
GND	Ground zero voltage potential.
S	Source terminal to the switch.
D	Drain terminal of the switch.
V_{ZAP}	Input test voltage.
IN	Address input to the switch.
T_A	Ambient temperature.
V_{CTE}	Charge transfer error.
V_{CT}	Crosstalk between channels.
V_{ISO}	Off isolation from source to drain of a closed switch.

6.5 Logistic support. Unless otherwise specified, microcircuits acquired for Government logistic support shall be acquired to device class S for National Aeronautics and Space Administration or class B for Department of Defense (see 1.2.2) and lead finish C (see 3.3). Longer length leads and lead forming shall not affect the part number.

6.6 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Military device type	Generic-industry type
01	200
02	201

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6.7 Handling. MOS devices must be handled with certain precautions to avoid damage due to accumulation of static charge. Input protection devices have been designed in the chip to minimize the effect of this static buildup. However, the following handling practices are recommended:

a. Devices should be handled on benches with conductive and grounded surfaces.

b. Ground test equipment and tools.

c. Do not handle devices by the leads.

d. Store devices in conductive foam or carriers.

e. Avoid use of plastic, rubber, or silk in MOS areas.

f. Maintain relative humidity above 50 percent, if practical.

6.8 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - AR, MI
Navy - OS, SH
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC

Agent:
DLA - ES

Preparing activity:
NASA - NA

(Project 5962-0430)

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